

Extending Scale BB from All-Cause to Cause-Specific Mortality: A Scenario Generator for Disease-Contingent Insurance under Economic-Value-Based Valuation

2026-07-31

Contents

§1 Introduction	2
1.1 Background — Two Turning Points	2
1.2 Approach and Novelty of This Study	3
1.3 Structure of the Findings and Plan of the Paper	3
§2 Related Work and Regulatory Requirements	4
2.1 SOA (2012) Scale BB and the Design Idea of the 2D-to-1D Reduction	5
2.2 Prior Research on APC Models and Disease-Rate Modeling	5
2.3 The Scenario Structure Required by Economic-Value-Based Regulation (ICS / IFRS 17 / Solvency II)	6
§3 Data and Methodology	7
3.1 Data and Target Diseases	8
3.2 The Scale BB Extension Algorithm	10
3.3 The APC Cohort-Penalty Extension	13
3.4 Baseline Methods and Evaluation Metrics	14
§4 Backtest Design	16
4.1 Validation framework — fit, project, validate	16
4.2 The 3 cutoff design and its purpose	16
4.3 Number of validation cells and statistical meaning	18
4.4 What the backtest measures (applying §3.4)	18
4.5 Reproduction	18
§5 Results I — Point-Forecast Accuracy	19
5.1 Point forecast accuracy of Scale BB-D alone ($y_c = 2014$)	19
5.2 Baseline comparison ($y_c = 2014$) — worse in all 24 cells	21
5.3 When the COVID-19 period is included in training ($y_c = 2021/2022$)	22
5.4 Summary — a negative conclusion and the turn to the next chapter	23
§6 Results II — The Directional Accuracy Finding	23
6.1 Directional accuracy at $y_c = 2014$ — 77.7–95.0% for the six main diseases	23
6.2 Across the 3 cutoffs — does the model capture direction regardless of the training period?	24
6.3 Baseline comparison — few methods carry direction information	25
6.4 What does the gap mean? — the essential difference between a structural model and a point predictor	25
6.5 Handling direction-reversal diseases — the scope and limits of calibration	26

§7 Repositioning — Scale BB-D as a Scenario Generator for ICS / IFRS 17	28
7.1 Reading the findings again — three structural properties	28
7.2 Implementation mapping to regulatory scenarios	29
7.3 Why the baseline methods cannot play the same role	29
7.4 Contrast with the ESG — a deterministic scenario generator for the disease risk axis	30
7.5 A preview of the adoption burden	30
7.6 Direct application to death benefits contingent on specific diseases — two layers of applicability	31
§8 Financial Impact Demonstration	31
8.1 Design of the Demonstration — A Simple Model Aligned with the ESR Implementation Rules	31
8.2 Scenarios and Model Points	32
8.3 Results — One Table of BEL Sensitivities	33
8.4 Discussion — Trend Shock versus Level Shock	34
8.5 Extension — Simple Calculation of Life Insurance Risk Capital and Insurance Liabilities	35
§9 Practical Implementation Guidelines	36
9.1 Guidance for Calibration and Tuning	36
9.2 Analysis of the Introduction Burden — Five Arguments and a Step-by-Step Path	37
9.3 A Real Case of Integration into an Existing System	38
§10 Limitations and Future Work	39
10.1 Proxy nature of the data — Limited scope of application and direct validation with incidence rates	39
10.2 Testability of long-term forecast accuracy — Handle it through scenarios, not by waiting for data	39
10.3 Building a systematic method for disease-specific calibration	40
10.4 Simplicity of the financial demonstration and re-running it on a practical model	40
10.5 Stochastic extension	40
§11 Conclusion	41

§1 Introduction

This paper extends Scale BB, the mortality improvement scale that the Society of Actuaries (SOA) published in 2012. We extend it from all-cause mortality to cause-specific (disease-specific) mortality. We test the extension with a 10-year backtest. After this test, we position the model as a scenario generator. Such a generator is required by economic-value-based valuation of insurance liabilities. Examples are the Insurance Capital Standard (ICS), the international accounting standard IFRS 17, and, in Japan, the economic-value-based solvency regulation (ESR). This chapter explains three things. First, the two changes in the environment that created this problem setting (§1.1). Second, the approach and the novelty of this study (§1.2). Third, the structure of the findings that runs through the paper, and the plan of each chapter (§1.3). A note on notation: we write Scale BB for the original SOA scale. We write Scale BB-D for the extended framework of this study (D stands for Disease, the extension to disease-specific rates).

1.1 Background — Two Turning Points

Structural change in disease rates. The COVID-19 pandemic was the largest shock to mortality and life expectancy in recent years. Its effect was not limited to the direct excess mortality in 2020–2022. It also reached the trends of cause-specific mortality and incidence rates. This happened through changes in care-seeking behavior, pressure on the healthcare system, and long-term health effects. OECD (2023) pointed out that the pandemic made mortality improvement assumptions unstable. It saw the main part of the shock as temporary, but noted large uncertainty about long-term health effects. It said that future improvement-rate models must be transparent and consistent with future expectations. A practice note by the American Academy of Actuaries (AAA) also gives guidance for setting assumptions after an external shock. It stresses the choice between static tables and generational tables, the adjustment of long-term improvement rates and the move

from short-term to long-term rates, and the importance of sensitivity analysis and documented judgment. In fact, our own backtest shows the same problem. The validation MAPE (Mean Absolute Percentage Error) of the same model changes by 6 to 8 times, depending on whether the COVID-19 shock is included in training (§5.3). Assumption practice in third-sector insurance (medical and disease coverage) has relied on static assumed rate tables. In this situation, the danger of the assumption that “past experience rates simply continue” became visible.

The turn in Japan’s capital regulation environment. In the same period, the valuation framework for insurance liabilities and capital is completing its move to an economic-value basis. The EU’s Solvency II applied from 2016. The international accounting standard IFRS 17 took effect in 2023. Then the Insurance Capital Standard (ICS) was adopted in December 2024. In Japan, the economic-value-based solvency regulation (ESR) started to apply from fiscal year 2025. This paper is being written exactly in the year when ICS takes effect and ESR applies. These frameworks share one point. The calculation of liabilities and capital is not a single best-estimate point forecast. It consists of a base scenario, a set of prescribed stress scenarios (level, trend, catastrophe), sensitivity disclosure, and ORSA (Own Risk and Solvency Assessment) stress (details in §2.3).

The gap on the disease-risk axis. At the crossing point of these two turns, there is a gap in practice. On the economic-risk axis, a systematic scenario supply tool exists: the Economic Scenario Generator (ESG). On the disease-risk axis, however, the main practice is to multiply a static assumed rate table by a uniform factor. This method cannot express a trend shock — a change with a time structure, expressed as an improvement rate. In other words, what practice needs now is not “a single point forecast with better accuracy”. It needs “a mechanism that generates several scenarios from the same base while keeping internal consistency”. This is the problem setting of this paper.

1.2 Approach and Novelty of This Study

Approach. Scale BB is a practical scale with a small number of parameters that are easy to interpret (§2.1). It smooths all-cause mortality experience in two dimensions, age by calendar year, and extracts improvement rates. It then blends the improvement rates of recent experience toward a long-term assumed improvement rate L by a convergence year P . This study extends this framework so that it can be applied to cause-specific mortality (§3.2–3.3). As input, we use cause-specific mortality from the Vital Statistics of Japan (MHLW) (1950–2024, 8 diseases x 3 sexes x 21 age groups). We test the model with a backtest that uses three different training cutoff years (§4). We check both point forecast accuracy and directional accuracy.

The novelty has two layers. First (Novelty A), the extension itself: applying a Scale BB type improvement scale to cause-specific mortality. The original Scale BB and its successors (the MP series and the CMI model) all target all-cause mortality only. As far as the authors know, no one has presented a systematic extension that separates and extracts, in one framework, the improvement trends, period shocks, and cohort effects that differ by cause of death (§2.1–2.2). Second (Novelty B), direct application to death benefits contingent on specific diseases. Some products pay benefits when death is caused by a specific disease. Examples are death coverage for the three major diseases, cancer death insurance, and disease-specific death riders by cause of death. For these products, the cause-specific mortality tested in this study is the assumption itself. The scenario generator can be applied directly. For disease incidence rates in medical insurance, the situation is different. Public statistics do not contain a long-term panel of incidence rates. So cause-specific mortality works as a proxy for validating the methodology (§3.1.3). This two-layer distinction — proxy and direct — is a premise that runs through the whole paper. Also, forecasting mortality after diagnosis (the case fatality rate) is outside the scope of this study. We make no claim about it in any form (§3.1.3, §10).

1.3 Structure of the Findings and Plan of the Paper

The structure of this paper is not a simple success story in which the original test plan worked as planned. It has a three-step structure: a negative conclusion, then an unexpected discovery, and finally new insight. We show the skeleton first.

Original goal and negative conclusion (§5). The original goal was to test whether the Scale BB-D extension is useful for point forecasts of disease rates. The answer is clearly no. Scale BB-D performs worse in MAPE

than the best simple baseline in all cutoffs and all 24 cells (8 diseases x 3 sexes). With training cutoff 2014, the gap is +16 to +44pp.

Unexpected discovery (§6). However, the picture changes completely when we measure the same model output by the “direction of change”. The directional accuracy for the six main diseases reaches 77.7–95.0%. This is far above the baseline methods, which carry direction information either not at all or only by chance. The model misses the point forecast but gets the direction right. This gap is not an accident. It follows from the design, which respects the long-term trend and deliberately ignores the noise at the end of the observation period (§6.4).

Translation into new insight (§7–§9). As §2.3 explains, what economic-value-based valuation mainly requires is not point forecast accuracy. It requires internal consistency between scenarios. The basic condition for this is that the direction of the improvement rates produced by the model has been verified in advance. From this viewpoint, §7 repositions Scale BB-D as a scenario generator. §8 quantifies the practical significance with a sensitivity demonstration of BEL (Best Estimate Liability). This demonstration follows the ESR implementation details (the risk-free interest rate curve and the stress factors set by regulatory notice). One contrast stands out (§8.4). The effect of the uniform level shock in the regulation is almost constant across entry ages (+17 to 19%). In contrast, the trend shock has an age gradient. For young entry ages, it reaches about 1.7 times the regulatory shock. This shows an age structure of risk that current practice with uniform factors cannot capture. §9 discusses calibration guidance for adoption and the adoption workload.

In one sentence, the claim of this paper is as follows. The Scale BB-D extension to cause-specific mortality performs worse than simple baselines in short-term point forecasts. But it explicitly outputs improvement rates $i^*(x, y)$ whose direction has been verified by backtest. So it has a structural advantage as the scenario generator that economic-value-based valuation requires. And cause-specific mortality plays two roles. For disease incidence rates in medical insurance, it is a proxy for validating the methodology. For death benefits contingent on specific diseases, it is the target itself. In the latter case, it can be applied directly, without the limits of a proxy.

The rest of the paper is organized as follows. §2 covers prior research and regulatory requirements. §3 covers data and methods. §4 covers the test design. §5 gives the results on point forecast accuracy. §6 presents the discovery about directional accuracy. §7 repositions the model as a scenario generator. §8 demonstrates the financial impact. §9 gives practical guidelines. §10 discusses limitations and future work. §11 concludes.

The next chapter first reviews the design ideas of the original Scale BB, our starting point. It then reviews the regulatory requirements that define the evaluation axis of this study — a structure that demands not a single point forecast but an internally consistent set of scenarios.

§2 Related Work and Regulatory Requirements

This chapter organizes two things. First, the prior research that is the starting point of this study. Second, the regulatory requirements that define the evaluation axes of this study. In §2.1 we review the design ideas of the original SOA (2012) *Mortality Improvement Scale BB*. The original work targets all-cause mortality. We contrast this with the cause-specific extension of this study (novelty A). In §2.2 we review prior research on APC (Age-Period-Cohort) models and disease-rate modeling. We identify the gap that this study fills. In §2.3 we organize the scenario structure that ICS, IFRS 17, and Solvency II require for the valuation of insurance liabilities. We place the regulatory requirements here, before the main part (§5 and later), for a clear reason. The evaluation axes of this study do not end with “point forecast accuracy”. What regulation mainly requires is not a single point forecast. It requires several scenarios with internal consistency. We want to confirm this before we start to interpret the results.

2.1 SOA (2012) Scale BB and the Design Idea of the 2D-to-1D Reduction

Scale BB is an alternative scale that the Society of Actuaries (SOA) presented in 2012. It was a response to the limits of the older one-dimensional mortality improvement scale, Scale AA. Its design has the following three elements.

1. Capturing the improvement structure in two dimensions. The scale takes the all-cause mortality experience of the U.S. Social Security Administration (SSA). It applies P-spline smoothing in two dimensions (2D), age by calendar year. It extracts the improvement rates as a surface. On the heat map, vertical patterns show period effects (example: influenza epidemics). Diagonal patterns show cohort effects (example: differences in smoking habits between generations). The idea of identifying these patterns and bringing period shocks into the model was made explicit here.
2. Blending toward a long-term improvement rate. The scale does not simply extrapolate the most recent observed improvement rates. Instead, it moves linearly from a fixed starting point (the year 2005) to a long-term assumed improvement rate of 1.0% per year at the convergence year (the cohort convergence period is 10 years; the blend function $h(y)$ in §7.4 of the original paper). This structure connects the recent trend and the long-term assumption in an explicit way. It follows the same idea as the UK CMI Mortality Projections Model, which blends from recently estimated improvement rates to a long-term assumed improvement rate.
3. A 1D reduction for compatibility with practice. The 2D information is reduced to a one-dimensional (1D) scale by age. The reduction is made so that the financial impact is approximately the same. This lets users apply the scale without changing their valuation software. Scale BB called itself an “interim scale”, but it was still widely used. A large reason is this design, which lowers the friction of adoption in practice.

Later, the SOA moved to a new system with the successor MP-2014 and its updates. In that system, a two-dimensional scale by age and calendar year is updated every year. So the framework of capturing the improvement structure in 2D has itself become the standard in U.S. practice.

Contrast with this study (novelty A). The original Scale BB and its successor scales all target all-cause mortality. For cause-specific (disease-specific) mortality, each cause has its own improvement trend, period shocks, and cohort effects. As far as the authors know, no systematic extension has been presented that separates and extracts these within one common framework. This study makes this extension (§3.2–3.3). It then connects the output to the scenario generation required by the regulations described in §2.3 (§7). The extended framework has a double applicability. It can serve as a proxy for method validation for disease incidence rates in medical insurance. It can also serve as a direct assumption for death benefits contingent on specific diseases. We define this in §3.1.3.

2.2 Prior Research on APC Models and Disease-Rate Modeling

Two lines of mortality modeling. There are, broadly, two lines of research for forecasting future mortality. The first line is stochastic models. It started with the log-bilinear model of Lee and Carter (1992). It was extended with a cohort term (Renshaw and Haberman 2006). The CBD model focused on old-age mortality (Cairns, Blake and Dowd 2006). Smoothing and extrapolation with P-splines followed (Currie, Durban and Eilers 2004). These models give future mortality as a probability distribution. The second line is practical scales. Examples are the Scale AA / Scale BB / MP series (U.S.) and the CMI model (UK). They are given as deterministic tables of improvement rates. They have a small number of parameters that people can interpret (long-term improvement rate, convergence period). They focus on a form that valuation practice can use directly. This study stands in the second line. It extends that all-cause framework to cause-specific rates. The two lines share the work in a clear way: stochastic models give distributions, and practical scales give deterministic scenarios. This matches the fact that the regulatory requirements in §2.3 are based on deterministic stresses. (A stochastic extension is a future task in §10.)

APC models. The decomposition into three effects — age, period, and cohort — was formalized in epidemiology by Holford (1983), Clayton and Schifflers (1987), and others. There is a rich body of work, mainly on descriptive analysis of cancer incidence and mortality. The identification problem comes from the linear dependence of the three effects (age + cohort = period). Its solution through constraints was also organized in this line of

research. The APC extension of this study (§3.3) follows Holford’s convention (reference-cell constraints, and moving the linear component into the period effect). However, the main goal of epidemiological APC research is to describe the past and to attribute causes. It does not provide a framework that turns the output into the assumption format of insurance practice — rate tables by age and calendar year with improvement scales, and future projections that valuation systems can consume.

Projection of disease rates. At the international level, large epidemiological projects such as the Global Burden of Disease (GBD) provide long-term estimates of disease-specific incidence and mortality. But their goal is to estimate the public health burden. They do not have the granularity and format that the valuation of insurance liabilities needs: consistency of age-specific rates as a surface, and a parameterization that allows scenario operations. On the insurance side, the development of improvement scales for incidence rates (morbidity) is far behind mortality. In Japan, one reason is a limit of official statistics: the Patient Survey (MHLW) runs on a 3-year cycle, so a long-term panel cannot be formed (§3.1). In Japanese practice, the Standard Mortality Tables and each company’s experience rate tables are static tables. For disease rates, no officially maintained improvement scale exists.

The gap this study fills. In short: (i) prior research on improvement scales is limited to all-cause mortality; (ii) cause-specific and disease-specific APC analysis stays at epidemiological description and is not provided in a practical assumption format; (iii) neither line presents the connection in which the direction of the improvement-rate output is first validated by backtest and then used to generate the scenarios that regulation requires. This study fills the crossing point of these three: the Scale BB extension to cause-specific rates (novelty A), output in a practice-compatible format, and use as a scenario generator based on directional validation (§6–§7).

2.3 The Scenario Structure Required by Economic-Value-Based Regulation (ICS / IFRS 17 / Solvency II)

A shift in the regulatory environment. The frameworks for valuing insurance liabilities and capital are in a period of transition. They are moving from static valuation under lock-in assumptions to an economic-value-based approach. In the EU, Solvency II was applied in 2016. In international accounting standards, IFRS 17 became effective in 2023. The IAIS adopted the Insurance Capital Standard (ICS) in December 2024. In Japan, the economic-value-based solvency regulation (ESR) started to apply from fiscal year 2025. The time of writing of this paper is exactly the year when ICS becomes effective and ESR starts to apply (§1).

These frameworks share one common point. The calculation of liabilities and capital is not built from “a single best estimate point forecast”. It is built from “a base scenario plus a set of prescribed stress scenarios”. The required scenarios can be organized into roughly three types.

- Level shock: an immediate, uniform change in the rate level (for example, a uniform increase or decrease of mortality)
- Trend shock: a change in the future improvement (or worsening) trend — an intervention in the time structure of the rates themselves
- Catastrophe shock: a temporary excess placed on one specific year only (for example, a pandemic)

On top of these, there is sensitivity disclosure (IFRS 17: disclosure of the change in liabilities when key insurance risk assumptions move within a reasonable range). There are also ORSA and medium- to long-term stress tests (a quantitative translation of narrative scenarios). The table below organizes the concrete calculations and the rate-scenario inputs they need.

Calculation	Required rate-scenario input
BEL (Best Estimate Liability)	Future rates of the base scenario $m(x, y)$
Risk adjustment (IFRS 17) / risk margin	Future rates stressed in the unfavorable direction
ICS capital charge for insurance risk	Future rates with level shock and trend shock applied
Catastrophe risk	Future rates with an excess shock placed on specific years only

Calculation	Required rate-scenario input
Sensitivity disclosure (IFRS 17)	Future rates with only one single assumption moved, and the resulting change in liabilities
ORSA / medium- to long-term stress	Future rates that translate narrative scenarios, such as worsening public health or a slowdown of medical progress

Internal consistency between scenarios. Each calculation in the table above is not an independent forecasting problem. It is a problem of generating a set of scenarios that start from the same base. Therefore the following properties are required.

1. All scenarios must keep the same age-curve shape (smoothness of the rate surface).
2. Only the rate level and the trend may move, while the relative relations between ages are kept.
3. It must be possible to put separate shocks on the short term and the long term (example: a trend shock for 5 years only, then normal afterwards).
4. When many scenarios run through the same calculation flow, the results must be comparable with each other.

If these break down, the difference between base and stress becomes a mix of two things: the effect of the stress itself, and the calibration error of the model. Then it cannot be used as a basis for regulatory capital calculation or disclosure. In particular, suppose the sign of the trend that the model outputs (improvement or worsening) is wrong in the first place. Then the operation of adding a trend shock loses its meaning. In other words, prior validation of the direction of the improvement rates is a basic condition for a scenario generator (we measure this in §6 and discuss it in §7).

A blank space in current practice. In current assumption practice for disease incidence rates (and cause-specific mortality), the main method is to multiply a static assumed rate table by a uniform safety loading factor. This method can express a level shock. But it cannot explicitly express a trend shock — a change with the time structure of an improvement rate. On the economic risk axis, an Economic Scenario Generator (ESG) provides a systematic supply of scenarios. On the disease risk axis, no corresponding scenario-generation mechanism exists today. (We discuss this contrast in detail in §7.)

Implication of this chapter. Once we confirm this structure of requirements first, we see that the validation in this study should be read along two axes. The first axis is “does the point forecast hit the target”. As we report in §5, on this axis our framework consistently performs worse than simple baselines. The second axis is “can it generate the scenario structure in the table above with internal consistency”. This is what regulation mainly requires. The process of turning the negative result on the first axis into a question on the second axis forms the narrative structure of this paper from §5 to §6–§7.

The regulatory requirements organized in this chapter become the coordinate axes for evaluating the point forecast results in §5. In §7 they are connected one-to-one with the output of our framework (the explicit output of the directionally validated improvement rates $i^(x, y)$). The next chapter, §3, defines the data and the extended algorithm used for the validation.*

§3 Data and Methodology

This chapter describes our framework, which extends Scale BB from all-cause mortality to cause-specific (disease-specific) mortality. We present the data used for the validation (§3.1), the mathematics of the extended algorithm (§3.2), the APC extension that separates cohort effects (§3.3), and the baseline methods and evaluation metrics used for comparison (§3.4). The goal of this chapter is to make the later chapters reproducible: the validation of point forecast accuracy (§5) and the findings on directional accuracy (§6). We define all notation here.

About notation: the math symbols follow the earlier study, the SOA (2012) *Mortality Improvement Scale BB Report*. Age is x , calendar year is y , the long-term improvement rate is L , the end (convergence) year of the blend is P , and the slope of the log-linear regression is s . We use the continuous piecewise-linear blend function $h(y)$ from §7.4 of the original report.

3.1 Data and Target Diseases

3.1.1 Input data

The input for the validation is the cause-specific crude mortality rate $m(x, y)$ per 100,000 total population. We took it from Table 5-15 of the Vital Statistics of Japan (MHLW), which gives annual trends by cause of death (cause of death $\times$ sex $\times$ 5-year age group $\times$ year). Here x is the age group and y is the calendar year.

Item	Content
Source	Vital Statistics Table 5-15 (annual-trend cause-of-death classification $\times$ sex $\times$ 5-year age group $\times$ year)
Period	1950–2024 (5-year steps for 1950–2010, yearly for 2013–2024)
Unit	Crude mortality rate per 100,000 population
Target diseases	8 diseases (table below)
Sex	3 groups: total / male / female
Age	21 groups from 0–4 to 100+ (the validation uses the 14 groups for ages 20–89)

Table 5-15 is a cause-of-death time series in which 5-year age groups $\times$ year are available over a long period. The observation years are not evenly spaced (5-year steps in the first half, yearly in the second half). The annualization logic in §3.2 handles this point (equation (3.3)). Under its terms of use, the Vital Statistics data can be used for commercial purposes.

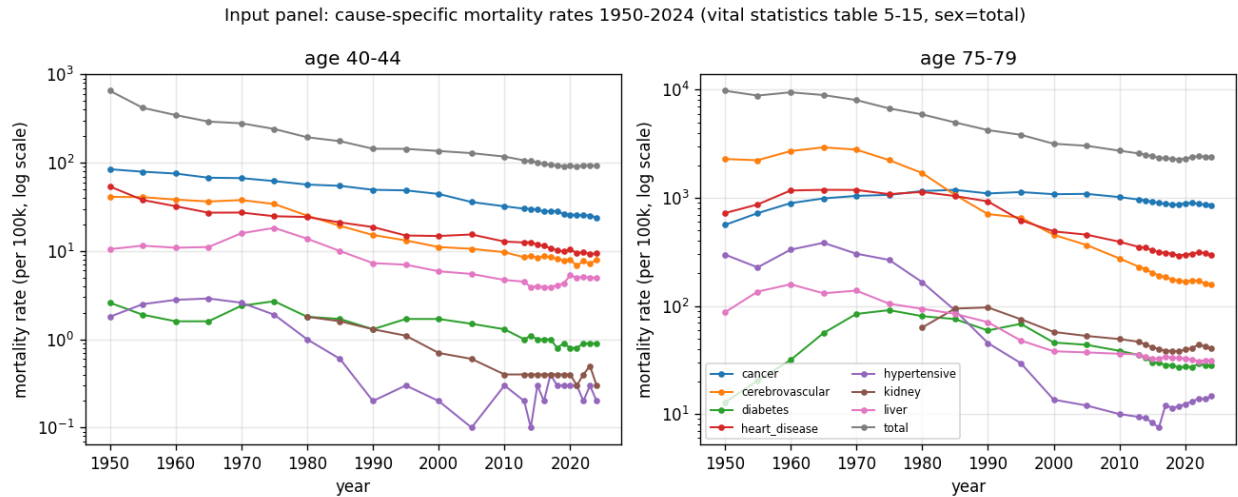


Figure 1: Overview of the input panel — cause-specific mortality trends for 8 diseases (sex=total, ages 40–44 / 75–79, log scale)

Figure 3.1: Overview of the input panel — the cause-specific mortality rates $m(x, y)$ from 1950 to 2024 (sex = total; left: ages 40–44; right: ages 75–79; log scale). The markers are observation points. They show the

uneven spacing: 5-year steps in the first half (1950–2010) and yearly points in the second half (2013–2024). The diseases differ greatly in level (a gap of more than 100 times) and in the shape of the long-term trend. For example, `cerebrovascular` falls steadily, while `cancer` rises and then turns down. (Generation script: `reproduction/backtest/make_paper_figures.py`)

3.1.2 Target disease mapping

From the definitions of insured diseases (`disease_estat_mapping.csv`), we built a panel of the 8 diseases for which Table 5-15 provides age-specific time series.

disease_id	5-15 cause code	Label	Remarks
<code>cancer</code>	Hi022017	Cancer (malignant neoplasms)	Mortality proxy for the disease incidence rate
<code>diabetes</code>	Hi03	Diabetes	Same as above
<code>hypertensive</code>	Hi042017	Hypertensive diseases	Same as above
<code>heart_disease</code>	Hi05	Heart disease (excluding hypertensive)	Same as above
<code>cerebrovascular</code>	Hi06	Cerebrovascular disease	Same as above
<code>liver</code>	Hi11	Liver disease	Same as above
<code>kidney</code>	Hi12	Kidney failure	Table 5-15 has no glomerular diseases etc., so kidney failure only
<code>total</code>	Hi00	Total	Sum of all causes

Ischemic heart disease (`heart_ischemic`) does not exist in the annual-trend cause-of-death classification of Table 5-15, so we excluded it. The panel shape is 8 diseases $\times$ 3 sexes $\times$ 25 years $\times$ 21 age groups = 12,600 rows.

3.1.3 The dual role of the data — proxy and direct target

The cause-specific mortality rates used in this study play two different roles, depending on the target product. This distinction is a core assumption. It defines the practical scope of the framework (§7 and §8) and its limits (§10).

1. For the disease incidence rates of medical insurance, the data are a “proxy” for validating the methodology. Medical insurance that pays benefits when a disease occurs really needs disease incidence rates as its assumption. However, as noted above, Japan’s public statistics have no long-term incidence panel by disease $\times$ age $\times$ year. So we use cause-specific mortality, which we expect to share the same age and cohort structure as incidence, as a proxy. In this way we validate the methodology of the Scale BB extension itself. Because it is a proxy, this validation cannot separate level differences that come from changes over time in the case fatality rate (mortality after diagnosis). We state this reservation clearly in §10.
2. For death benefits contingent on specific diseases, the data are the “direct assumption” itself. Some products pay death benefits based on the cause of death: death coverage for the three major diseases, cancer death insurance, and cause-specific disease-death riders. For these products, cause-specific mortality is not a proxy. It is the assumption itself for computing the present value of benefits. We apply the framework directly to these products.

Note: The data in this study are cause-specific mortality rates per total Japanese population. They are not mortality after diagnosis (case fatality rate = deaths by cause $\div$ number of diagnosed cases). Case fatality rates need observed incidence rates in the denominator, so they are outside the scope of this study. We do not try to “predict the prognosis after diagnosis” (see §10).

3.2 The Scale BB Extension Algorithm

Scale BB here extends SOA (2012) *Mortality Improvement Scale BB* to disease-specific mortality. It has two phases: Phase 1 (smoothing of observations and extraction of improvement rates) and Phase 2 (blending with the long-term rate and future projection). The implementation is based on the pure functions in `_scalebb_core/model.py` (`fit_scale_bb` / `project_scale_bb`).

3.2.1 Phase 1 — Two-dimensional smoothing and improvement-rate extraction

We take the log of the observed crude rate $m(x, y)$ and set $z(x, y) = \log m(x, y)$. Cells that are 0 or below, or missing, receive weight 0.

We apply two-dimensional Whittaker-Henderson penalized smoothing to the log-rate matrix $Z = (z_{x,y})$. This is equivalent to the P-spline smoothing of the original SOA report (tensor-product B-spline + difference penalty). We find the smoothed matrix $\hat{Z}$ that minimizes the following objective function.

$$J(\hat{Z}) = \sum_{x,y} w_{x,y} (z_{x,y} - \hat{z}_{x,y})^2 + \lambda_{\text{age}} \|D_{\text{age}}^{(d)} \hat{Z}\|_F^2 + \lambda_{\text{year}} \|\hat{Z} (D_{\text{year}}^{(d)})^\top\|_F^2 \quad (3.1)$$

Here,

- $w_{x,y}$ are the observation weights (missing and non-positive cells get 0),
- $D^{(d)}$ is the difference matrix of order d (default $d = 2$),
- $\lambda_{\text{age}}, \lambda_{\text{year}}$ are the smoothing strengths in the age direction and the calendar-year direction,
- $\|\cdot\|_F$ is the Frobenius norm.

If we arrange $\text{vec}(\hat{Z})$ in column-major (Fortran) order, the stationary condition of equation (3.1) becomes a sparse linear system. Set $P_{\text{age}} = I_{n_{\text{year}}} \otimes (D_{\text{age}}^{(d)\top} D_{\text{age}}^{(d)})$, $P_{\text{year}} = (D_{\text{year}}^{(d)\top} D_{\text{year}}^{(d)}) \otimes I_{n_{\text{age}}}$, and $W = \text{diag}(\text{vec}(w))$. Then we solve

$$(W + \lambda_{\text{age}} P_{\text{age}} + \lambda_{\text{year}} P_{\text{year}}) \text{vec}(\hat{Z}) = W \text{vec}(Z) \quad (3.2)$$

with a sparse matrix solver. On a coarse age $\times$ year grid (at most about 80×80), it converges instantly. We recover the smoothed rates as $\tilde{m}(x, y) = \exp(\hat{z}(x, y))$. Figure 3.2 compares the rates before and after smoothing on real data (heart disease, sex = total).

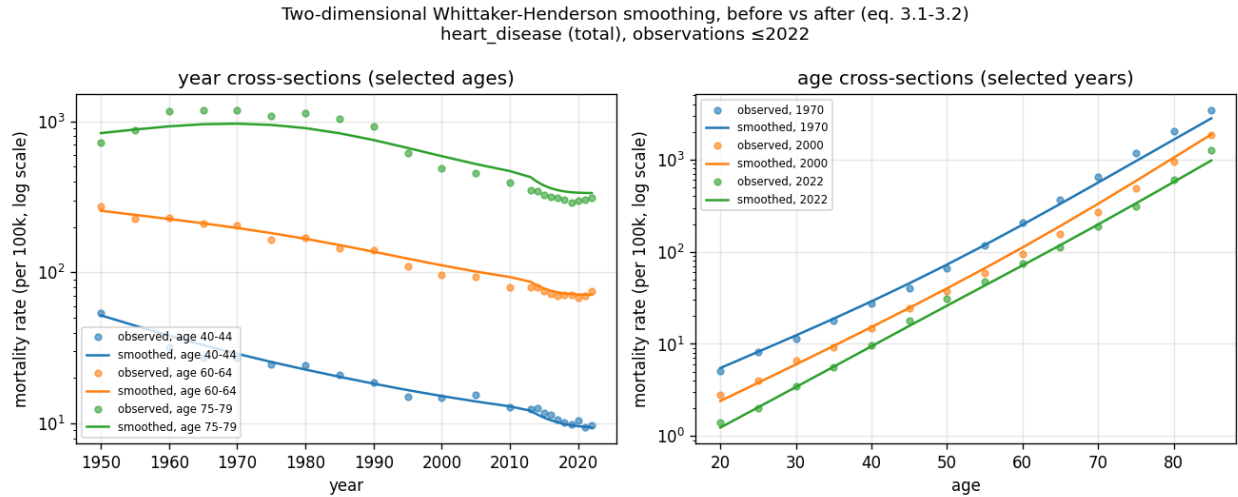


Figure 2: Before and after smoothing — observed rates and two-dimensional Whittaker-Henderson smoothed rates for heart disease, total (left: cross-sections along calendar year, right: cross-sections along age)

Figure 3.2: Comparison before and after two-dimensional Whittaker-Henderson smoothing (equations 3.1–3.2, $\lambda_{\text{age}} = \lambda_{\text{year}} = 40$) (heart disease, sex = total, observations ≤ 2022 , log scale). The markers show the observed crude rates $m(x, y)$; the solid lines show the smoothed rates $\tilde{m}(x, y)$. The left panel shows cross-sections along calendar year (3 representative ages). The short-term noise typical of yearly data is smoothed out, while the long-term trend is kept. The right panel shows cross-sections along age (3 representative calendar years). The age slope, which is almost linear on the log scale, is recovered smoothly. The smoothing works in the age and calendar-year directions at the same time. So it has a consistency that one-dimensional smoothing of each cross-section cannot give. (Generation script: `reproduction/backtest/make_paper_figures.py`)

Note that in the left panel, for ages 75–79, the smoothed rate stays a little below the peak of the observations around 1960–1990. The reason is that the default smoothing strength ($\lambda = 40$) prefers extracting the long-term trend over following the local level. This behavior is intended (the settings are the same for all diseases, §3.2.3).

Next, we extract the annual observed improvement rate $i(x, y)$ from the smoothed rates. The BFL (best-fit log-linear) method in §3.1 of the original report gets the annual improvement $1 - e^s$ from the slope s . We follow the same idea. Even when the observation years are unevenly spaced, we convert to the geometric mean improvement rate between two neighboring time points. Using the year gap $\Delta y_k = y_k - y_{k-1}$, we define

$$i(x, y_k) = 1 - \left(\frac{\tilde{m}(x, y_k)}{\tilde{m}(x, y_{k-1})} \right)^{1/\Delta y_k} \quad (3.3)$$

Here $i > 0$ means the rate falls (improvement), and $i < 0$ means the rate rises (worsening). The stability of the sign of this improvement rate is what the directional accuracy in §6 measures, and it leads to the scenario generation in §7.

3.2.2 Phase 2 — Long-term rate blending and future projection

In Phase 2, we blend the smoothed improvement rate at the last observed year, $i(x, y_{\text{obs}})$, linearly toward the long-term assumed improvement rate L (default 1%) up to the convergence year P (default 2035). At high ages, improvement slows down and disappears in reality. To reflect this, we introduce an age-specific taper factor $\tau(x)$ and set the age-specific long-term rate to $L_{\text{age}}(x) = L \cdot \tau(x)$. With a taper start age a_0 (default 90) and a taper end age a_1 (default 120), we define

$$\tau(x) = \begin{cases} 1 & x \leq a_0 \\ 1 - \frac{x - a_0}{a_1 - a_0} & a_0 < x < a_1 \\ 0 & x \geq a_1 \end{cases} \quad (3.4)$$

(This matches §5.2 of the original report: the long-term rate applies up to age 90 and falls linearly to 0% at age 120.)

The weight for the linear transition from the last observation to the long-term rate follows the continuous piecewise-linear blend function $h(y)$ in §7.4 of the original report. The original report defines $h(y)$ as a multiplicative factor that moves linearly from 1 to L between a fixed start year (2005) and P . We change this in two ways: (i) we generalize the start year to the last observed year y_{obs} (to allow a variable cutoff and the extension to causes of death); (ii) we use $h(y)$ not as a multiplicative rescaling of the rates, but as the weight of a linear interpolation between the observed improvement rate and the long-term rate. That is, we set

$$h(y) = \begin{cases} 0 & y \leq y_{\text{obs}} \\ \alpha_y = \frac{y - y_{\text{obs}}}{P - y_{\text{obs}}} & y_{\text{obs}} < y < P \\ 1 & y \geq P \end{cases}$$

Then the final improvement rate $i^*(x, y)$ is given by

$$i^*(x, y) = (1 - h(y)) i(x, y_{\text{obs}}) + h(y) L_{\text{age}}(x) \quad (3.5)$$

In other words, in the observed period ($y \leq y_{\text{obs}}$) the rate keeps the value at the last observed year, $i(x, y_{\text{obs}})$. It then moves linearly toward the long-term rate $L_{\text{age}}(x)$ up to the convergence year P . From P on ($y \geq P$), it stays flat at $L_{\text{age}}(x)$. Figure 3.3 shows a real example of $i^*(x, y)$ on real data (heart disease, sex = total, $y_{\text{obs}} = 2022$).

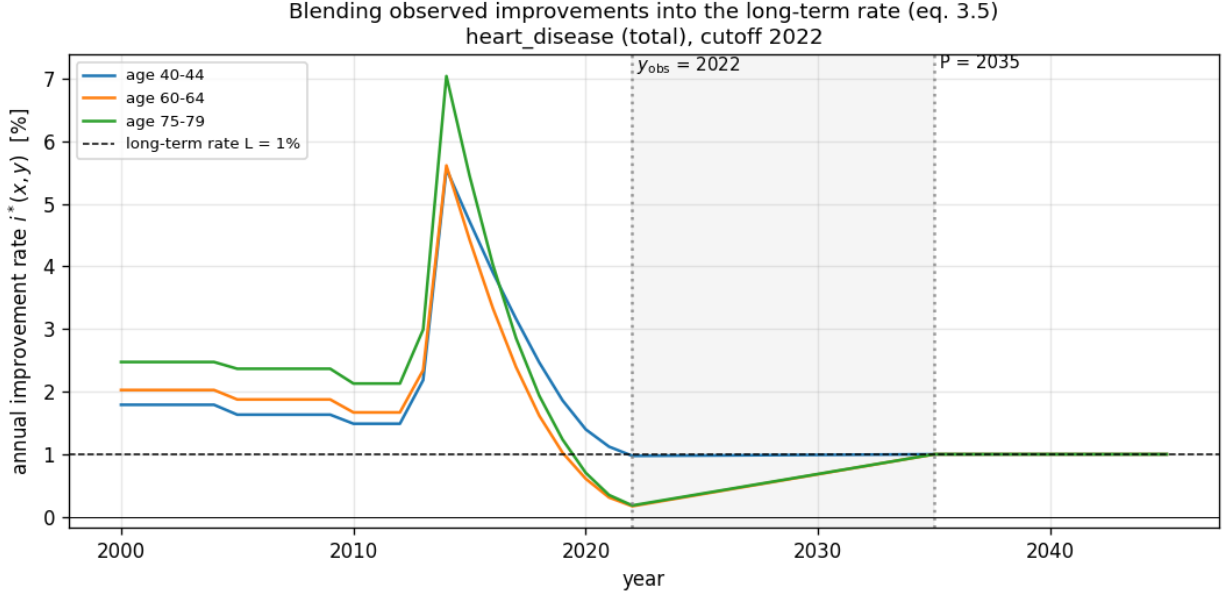


Figure 3: Example of the improvement-rate blend — annual improvement rate $i^*(x, y)$ for heart disease, total, cutoff 2022

Figure 3.3: A real example of the blend in equation (3.5) (heart disease, sex = total, $y_{\text{obs}} = 2022$, 3 representative ages). In the observed period, the lines show the observed improvement rates smoothed in Phase 1 (including the drop in improvement during the COVID-19 period). From the value $i(x, y_{\text{obs}})$ at the last observed year $y_{\text{obs}} = 2022$, the rate moves linearly to the long-term rate $L = 1\%$ up to the convergence year $P = 2035$ (shaded band), and stays flat after that. (Generation script: `reproduction/backtest/make_paper_figures.py`)

We add three comments on Figure 3.3. First, the sharp peak in 2014–2015 (5.6 to 7.0% per year) shows the smoothed curve capturing the fast fall of heart disease mortality in the mid-2010s. However, it is also affected by the annualization (equation 3.3) of the 2010→2013 gap, where the observation grid switches from 5-year steps to yearly data. So we do not read it as the true level of a single year. Second, the improvement rate at the last observed year 2022 is a little under 0.2% for ages 60–64 / 75–79. This reflects the stalled improvement in the COVID-19 period, and it is far below the historical level (1 to 2%). The blend starts from this reduced value. It does not assume that the recent anomaly lasts forever, and it does not assume an immediate return to normal. Instead, it moves back to the long-term rate over the years up to the convergence year. Third, for ages 40–44, $i(x, 2022) \approx 1.0\%$ happens to be almost equal to L , so the blend section is almost flat. This shows that the blend really works only when the improvement rate at the last observation differs from the long-term rate.

We generate the future rates forward from the rate $m(x, y_0)$ of the base year y_0 , by accumulating the improvement rates.

$$m(x, y) = m(x, y_0) \prod_{u=y_0+1}^y (1 - i^*(x, u)) \quad (3.6)$$

The structural key point of the framework is this: the future rates are fully rebuilt as “base-year rate $\times$ cumulative $(1 - i^*)$ ”. If we replace $i^*(x, y)$, we get new future rates at once. So we can generate several scenarios with different long-term rates L or convergence years P in an internally consistent way, while keeping the same age-curve shape and the same base-year rates (§7).

3.2.3 Hyperparameters

The output of Scale BB depends on quantities estimated from the data, such as the smoothed matrix $\hat{Z}$ and the improvement rates $i(x, y)$. It also depends on settings that the analyst gives from outside, before the estimation. In this paper we call the latter hyperparameters. They are: $\lambda_{\text{age}}, \lambda_{\text{year}}, d$, which set the strength and smoothness of the smoothing (equation 3.1); L, P , which set the long-term improvement assumption (equation 3.5); and a_0, a_1 , which set the fade-out of improvement at high ages (equation 3.4). The table below lists them with their default values.

In this validation we did no disease-specific tuning. We applied the default preset of the experience-rate aggregation process (`config.yaml > scalebb_presets > defaults`) to all diseases. We discuss guidance for disease-specific calibration separately in §9.

Symbol	Parameter	Default
L	Long-term improvement rate (<code>long_term_rate</code>)	0.01 (1%)
P	Convergence year (<code>convergence_year</code>)	2035
λ_{age}	Smoothing in the age direction (<code>lam_row</code>)	40
λ_{year}	Smoothing in the calendar-year direction (<code>lam_col</code>)	40
d	Difference order (<code>diff_order</code>)	2
a_0, a_1	Age taper start and end	90, 120
y_{obs}	Last observed year (<code>last_observed_year</code>)	cutoff (§4)

Note on the range of settings: The table above shows the settings used for the backtest in this chapter (validation of point forecasts and direction). It corresponds to the reproduction package `reproduction/backtest/`. The forward-looking pipeline that generates generational assumed rate tables (`reproduction/generational/`, details in `reproduction/generational/README.md`) uses the age20 preset with `lam_col=60` (stronger smoothing in the calendar-year direction) and the age range `age_min=20`. This reduces year-to-year swings in the rates for young ages (starting at age 20). The core parameters other than λ_{year} ($L = 0.01, P = 2035, \lambda_{\text{age}} = 40, d = 2$) are the same in both.

3.3 The APC Cohort-Penalty Extension

The Age-Period (AP) model above absorbs the cohort effect $\gamma(c)$ (where $c = y - x$ is the birth cohort) into the period effect $\beta(y)$. So it cannot mathematically identify “long-term health effects specific to a cohort that experienced a pandemic at a certain age”. To separate this, we extend to an Age-Period-Cohort (APC) model with an additive decomposition (`scale_bb_apc_model.py`).

The observation model is

$$\log m(x, y) = \alpha(x) + \beta(y) + \gamma(c) + \varepsilon(x, y), \quad c = y - x \quad (3.7)$$

Here $\alpha(x)$ is the age effect. $\beta(y)$ is the period effect (health-system reforms, epidemics, the COVID-19 shock). $\gamma(c)$ is the cohort effect (lifestyle, access to medical technology, disease exposure in youth). The smoothing objective adds a diagonal (cohort) difference penalty to equation (3.1):

$$J(\hat{Z}) = \sum_{x,y} w_{x,y} (z_{x,y} - \hat{z}_{x,y})^2 + \lambda_{\text{age}} \|D_{\text{age}}^{(d)} \hat{Z}\|_F^2 + \lambda_{\text{year}} \|\hat{Z} (D_{\text{year}}^{(d)})^\top\|_F^2 + \lambda_{\text{cohort}} \|D_{\text{cohort}}^{(d)} \text{vec}(\hat{Z})\|^2 \quad (3.8)$$

We build the diagonal difference matrix $D_{\text{cohort}}^{(d)}$ as a second-order difference along the same cohort. On $\text{vec}(\hat{Z})$ (column-major), it takes the form $\hat{z}_{i+2,j+2} - 2\hat{z}_{i+1,j+1} + \hat{z}_{i,j}$ (here i, j are the matrix indexes for age and calendar year). After smoothing, we extract α, β, γ with an iterative weighted least-squares method (`decompose_apc_additive`).

Identifiability. Because of the linear dependence $\text{age} + \text{cohort} = \text{period}$, an APC model cannot identify the absolute levels and the linear trend parts of the three effects. Our implementation handles this with two tools used together.

- Second-order difference penalty ($d = 2$): the constant and linear parts are not penalized. Under the penalty, only the nonlinear parts of order 2 and higher are identified uniquely. Our research interest is the nonlinear deviation of cohorts that were at a certain age during the pandemic. This nonlinear part is identifiable, so it matches the research hypothesis.
- Reference cell constraints: we impose $\alpha(x_{\min}) = \beta(y_{\min}) = 0$ to remove the indeterminacy of the constant level (Holford 1983).

We merge the linear part into the period effect, following the Holford convention, and report results as relative improvement rates and residuals.

Treatment of the COVID-19 period. The pandemic period (2020–2022) is a structural break outside the normal trend. Simple smoothing would distort the estimate of the long-term improvement rate. Our implementation offers 3 modes.

Mode	Treatment	Recommended use
<code>weight_down</code>	Reduce the observation weights of the COVID years: $w \mapsto c_w \cdot w$ ($c_w \in [0, 1]$, default 0.3)	Diseases whose long-term improvement continues smoothly (cancer)
<code>dummy</code>	Replace $\beta(y)$ in the COVID years with $\beta_{\text{corr}}(y)$, the linear interpolation from non-COVID years, and rebuild the rates with β_{corr} . Keep the shock part $\beta(y) - \beta_{\text{corr}}(y)$ separately	Diseases with clear irregular movements in the COVID period (heart disease, cerebrovascular disease)
<code>none</code>	COVID years also keep weight 1.0 (reference for comparison)	Baseline

For the projection, we must extrapolate $\gamma(c)$ for new, unobserved cohorts that appear during the projection period. The default is `last_drift`, which extends the last first-order difference (consistent with the Scale BB philosophy). A conservative option is `flat`, which holds the last value fixed.

3.4 Baseline Methods and Evaluation Metrics

3.4.1 Three baseline methods

To evaluate the usefulness of Scale BB in relative terms, we compare it with three non-Scale-BB baselines. They either have no explicit concept of an improvement rate, or hold it in a different way. All of them are built for each age x separately. Let y_c be the cutoff year (the end of the training period).

Method	Forecast rule	Intuition
naive_last	$\hat{m}(x, y) = m(x, y_c)$	“The latest value continues.” By construction, $\hat{m}$ does not depend on y
mean_3pts	$\hat{m}(x, y) = \frac{1}{3} \sum_{k=0}^2 m(x, y_c^{(k)})$ (mean of the last 3 observation points)	Noise removal, level averaging
loglin_trend	Fit $\log m(x, y) = a_x + s_x y$ by OLS (ordinary least squares) for each age over the last 15 years, and extrapolate with $\hat{m}(x, y) = \exp(a_x + s_x y)$	A classical actuarial method. It has the same form as the BFL in §3.1 of the original report, and it implies an annual improvement of $1 - e^{s_x}$

naive_last carries the cutoff-year value forward. So its predicted change, defined below, is always 0 (it has no direction information). **loglin_trend** is the only baseline with explicit direction information, through the age-specific slope s_x . So it is the most meaningful rival of Scale BB (§6).

3.4.2 Evaluation metrics

For each validation cell (age $x \times$ validation year $y \times$ disease $\times$ sex), we compare the actual value $m_{\text{act}}(x, y)$ with the forecast $\hat{m}(x, y)$ and compute the metrics below. Cells with an actual value of 0 are excluded from MAPE.

Point forecast accuracy metrics. For the set of validation cells $\mathcal{C}$ (with $N = |\mathcal{C}|$ elements),

$$\text{MAPE} = \frac{100}{N} \sum_{(x,y) \in \mathcal{C}} \frac{|\hat{m}(x, y) - m_{\text{act}}(x, y)|}{m_{\text{act}}(x, y)} \quad [\%] \quad (3.9)$$

$$\text{bias} = \frac{1}{N} \sum_{(x,y) \in \mathcal{C}} (\hat{m}(x, y) - m_{\text{act}}(x, y)) \quad [\text{per } 10^5] \quad (3.10)$$

A positive bias means the forecast is too high; a negative bias means it is too low. As support, we also use RMSE (Root Mean Square Error) and the mean relative bias ($\frac{100}{N} \sum (\hat{m} - m_{\text{act}})/m_{\text{act}}$).

Directional accuracy (DA below). This metric supports the central finding of this study (§6). It measures how well the forecast predicts whether the rate moves up or down, compared with the last training year y_c . For each cell, we define the actual change and the predicted change as

$$\Delta_{\text{act}}(x, y) = m_{\text{act}}(x, y) - m(x, y_c), \quad \Delta_{\text{pred}}(x, y) = \hat{m}(x, y) - m(x, y_c) \quad (3.11)$$

We restrict to the set of cells where the actual value changed, $\mathcal{D} = \{(x, y) : \Delta_{\text{act}}(x, y) \neq 0\}$, and compute the sign agreement rate

$$\text{DA} = \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} \mathbf{1}[\text{sign } \Delta_{\text{pred}}(x, y) = \text{sign } \Delta_{\text{act}}(x, y)] \quad [\%] \quad (3.12)$$

Here $\mathbf{1}[\cdot]$ is the indicator function.

Under this definition, **naive_last** has $\Delta_{\text{pred}} \equiv 0$ by construction and carries no direction information (its DA is in effect 0). This is an intended axis of evaluation: DA asks whether the model tried to call the direction. Point forecast accuracy (MAPE) measures “getting the level right”. Directional accuracy (DA) measures “getting the sign of the trend right”. The gap between the two is the starting point of the argument for repositioning Scale BB as a scenario generator, not a point forecaster (§7).

The notation of this chapter is used throughout §4 and later. §4 gives the details of the validation design (3 cutoffs). §5 gives the point forecast accuracy results. §6 gives the directional accuracy results. §10 discusses head-on the limits that come from the proxy nature of the data.

§4 Backtest Design

This chapter describes the design of the backtest. The backtest uses the model and the evaluation metrics defined in §3. The central design choice is the “3 cutoff design”. We set the training cutoff year at three points: 2014, 2021, and 2022. This lets us separate two cases. In one case, COVID-19 (the structural break of 2020–2022) reaches the model as an unknown disturbance. In the other case, the model receives it as a learned shock. This design is the base for reading both the point forecast accuracy results (§5) and the directional accuracy findings (§6), including how they depend on the training period. We follow the notation of §3. The calendar year is y , and the cutoff year is y_c .

4.1 Validation framework — fit, project, validate

For each cutoff year y_c , we take the following steps.

1. Fit: We fit Scale BB-D (§3.2) and the 3 baselines (§3.4.1) to the cause-specific mortality rates $m(x, y)$ from the first observation year $y_0 = 1950$ to y_c . All methods use the same training period. This makes the comparison between methods fair.
2. Project: Starting from the last observed year $y_c (= y_{\text{obs}})$, we generate future rates $\hat{m}(x, y)$ up to the end of the validation period $T = 2024$ (Equations 3.5–3.6).
3. Validate: We compare the forecasts for the hold-out interval $y \in [y_c + 1, T]$ with the actual data. We compute the metrics of §3.4.2 (MAPE, bias, and directional accuracy DA) for each cell of age $\times$ validation year.

The validation covers 8 diseases $\times$ 3 sexes. Ages run from 20 to 89, in 14 five-year age groups. The training period is a mixed series with unequal spacing: 5-year steps (1950–2010) and yearly steps (2013– y_c). However, the improvement rates stay consistent because Equation (3.3) converts them to annual rates.

Note that the convergence year of Scale BB-D is $P = 2035$ (§3.2.3). This year lies after every validation interval. So for every cutoff, the validation interval $[y_c + 1, 2024]$ sits fully inside the blending interval $y_{\text{obs}} < y < P$. Therefore, the forecasts in the validation interval reflect the linear blend of the improvement rate at the last observed year $i(x, y_c)$ and the long-term rate $L_{\text{age}}(x)$ (Equation 3.5, the region where $0 < h(y) < 1$).

4.2 The 3 cutoff design and its purpose

We set the training cutoff at three points. In this way, we switch step by step whether the pandemic period is part of the model’s training or not.

cutoff y_c	Training period	Training observation years	Validation period	Forecast horizon	How COVID-19 enters
2014	1950–2014	15	2015–2024	10 years ahead	Unknown disturbance (the pandemic is not in the training data)

cutoff y_c	Training period	Training observation years	Validation period	Forecast horizon	How COVID-19 enters
2021	1950–2021	22	2022–2024	3 years ahead	Learned shock (the pandemic period is in the training data)
2022	1950–2022	23	2023–2024	2 years ahead	Includes 1 post-pandemic year; uses the most recent data as much as possible

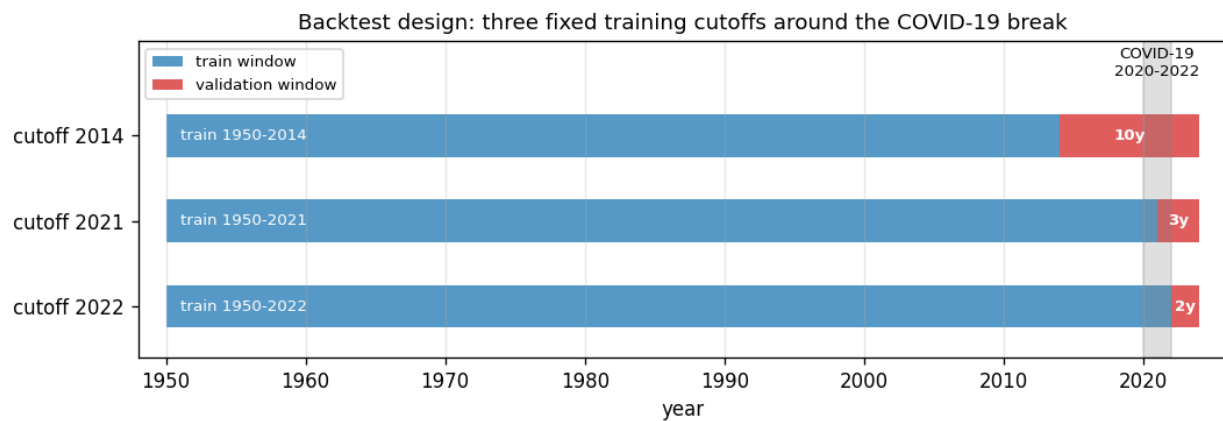


Figure 4: Diagram of the 3 cutoff design — the training and validation periods of each cutoff and the position of the COVID-19 period

Figure 4.1: Diagram of the 3 cutoff design. Each row shows one cutoff. Blue is the training period ($1950-y_c$, “train window” in the figure). Red is the validation period ($y_c + 1-2024$, “validation window” in the figure). The shaded area is the COVID-19 structural break period (2020–2022). For cutoff = 2014, the break falls in the validation period (unknown disturbance). For cutoff = 2021 / 2022, the break falls in the training period (learned shock). (Generated by: `reproduction/backtest/make_paper_figures.py`)

The three cutoffs have the following roles.

- cutoff = 2014 — COVID enters as an unknown disturbance. Training stops at 2014, and the model forecasts 10 years ahead, 2015–2024. The excess mortality of 2020–2022 is completely unknown to the model. This is the hardest test of how fragile a trend-extrapolation model is against a structural break. The validation interval is the largest (10 years $\times$ 14 ages), so the estimate of directional accuracy is the most stable in a statistical sense (§4.3).
- cutoff = 2021 — COVID enters as a learned shock. The training data includes the pandemic period (2020–2021), and the model forecasts the stable post-COVID period (2022–2024). This measures how much the point forecast recovers when the structural change of the pandemic is part of the training (the MAPE reduction in §5).
- cutoff = 2022 — Use the most recent data as much as possible. Training includes the post-pandemic year 2022, and the model forecasts 2023–2024. This checks the upper limit of point forecast performance when the training period is as long as possible.

These three points are not a continuous rolling-origin scheme (moving the origin one step at a time). They are three fixed origins, chosen on purpose so that they sit around the structural break of the pandemic. We place the cutoffs “just before the break (2014)”, “at the end of the break (2021)”, and “after the break (2022)”. In this way, we can read the model’s response to the same structural break as a function of how much of the break is in the training data.

4.3 Number of validation cells and statistical meaning

The number of validation cells for each cutoff comes from 8 diseases $\times$ 3 sexes $\times$ 14 age groups $\times$ the number of validation years.

$$N_{\text{cell}} = \underbrace{8}_{\text{diseases}} \times \underbrace{3}_{\text{sexes}} \times \underbrace{14}_{\text{age groups}} \times \underbrace{H}_{\text{validation years}}$$

cutoff	Validation years H	Validation cells N_{cell}
2014	10	3,360
2021	3	1,008
2022	2	672

cutoff = 2014 has the largest number of cells. So its aggregate values by disease and by age are the most stable. In contrast, cutoff = 2021 / 2022 have short validation periods and few cells. Because of this, the metrics can vary a lot between diseases, and readers should keep this in mind. (This is why, in §6, the directional accuracy for cutoff = 2021 swings widely from disease to disease.)

Note that the directional accuracy DA (Equations 3.11–3.12) is evaluated only on the cells where an actual change happened, $\mathcal{D} = \{(x, y) : \Delta_{\text{act}} \neq 0\}$. So its effective number of cells is smaller than N_{cell} in the table above, and it differs by cutoff and by disease. The tables in §6 state the effective cell counts for each aggregate.

4.4 What the backtest measures (applying §3.4)

For each cutoff, we apply the two groups of metrics defined in §3.4.2.

- Point forecast accuracy (reported in §5): MAPE (Equation 3.9) and bias (Equation 3.10). These measure how close the forecast level is to the actual data.
- Directional accuracy (reported in §6): DA (Equation 3.12). This measures whether the model can predict if the rate moves up or down compared with the last observed year y_c .

We run Scale BB-D and the 3 baselines on the same 3 cutoff design and compare both groups of metrics. In this way, we quantify the gap “the model performs worse in point forecasts but predicts the direction correctly” (the turn from §5 to §6), including how it depends on the training period.

4.5 Reproduction

The script `run_all.sh` in the reproduction package `reproduction/backtest/` runs the full 3 cutoff design of this chapter in one go. The results for cutoff = 2014 go directly under `output/`. The results for cutoff = 2021 / 2022 go to `output/cutoff_2021/` and `output/cutoff_2022/`. Then `compare_cutoffs.py` produces the cross-cutoff comparison tables (matching the tables of §5.2–5.3 and Figures 5.4–5.5). The training and validation periods of each cutoff are set explicitly by the CLI arguments `--train-cutoff` / `--validation-end` (see the reproduction steps in §3).

Based on the 3 cutoff design set in this chapter, the next chapter §5 first reports the point forecast accuracy results. There we find that “Scale BB-D has a worse MAPE than the baselines at all cutoffs”. In the light of the design purpose of §4.2, this result is not a failure. It is the starting point that turns the question toward the directional accuracy findings of §6.

§5 Results I — Point-Forecast Accuracy

This chapter reports the point forecast accuracy results based on the 3-cutoff design of §4. We state the conclusion first. Scale BB-D performs worse than the best baseline in MAPE (Equation 3.9) at every cutoff and in every cell. Within the original goal (testing the usefulness of Scale BB-D for point forecasts), this result leads to a negative conclusion: “Scale BB-D is not suitable for short-term point forecasts.” This chapter reports this result openly and directly. It also quantifies how accuracy changes when COVID-19 is included in the training data or not. This is a by-product with independent value. All numbers below match the regenerated results in the reproduction package `reproduction/backtest/output/` (sex = total).

5.1 Point forecast accuracy of Scale BB-D alone ($y_c = 2014$)

We cut off the observations at 2014 and forecast the 10 years from 2015 to 2024. The table shows the accuracy of Scale BB-D alone.

Disease	MAPE [%]	bias [per 10^5]	MAPE 2015 (1 year ahead)	MAPE 2024 (10 years ahead)
cancer	22.41	+42.6	11.96	30.95
total	26.01	−496.2	10.99	41.54
heart_disease	31.27	−85.3	11.12	48.52
diabetes	35.59	−3.9	10.48	55.75
kidney	40.87	−5.1	13.46	59.39
cerebrovascular	47.14	−59.1	22.77	64.03
liver	49.49	−7.8	20.53	67.50
hypertensive	73.83	−7.1	37.02	88.33

There are three main observations.

1. The error grows as the forecast horizon gets longer. One year ahead (2015), the MAPE is 10.5–37.0%. This is an acceptable level for short-term forecasts in practice. But 10 years ahead (2024), it grows to 31.0–88.3%. Scale BB-D is a structural model. It respects the long-term trend, not the local level near the end of the observations. So when the horizon gets longer, the error from trend extrapolation builds up.
2. All diseases except **cancer** show a negative bias (the forecast is too low). A negative bias means forecast < actual. In other words, Scale BB-D extrapolated the continued improvement too far. It could not capture the excess mortality caused by COVID-19 in 2020–2022. This shows exactly the design hypothesis of §4.2: trend-extrapolation models are weak against structural breaks.
3. Only **cancer** shows a positive bias (the forecast is too high). The reason is that real progress in treatment was faster than the slow past decline. So the improvement in Scale BB-D was too small compared with the actual results.

These three points match Figures 5.1–5.3.

Figure 5.1: Yearly MAPE (left) and mean relative bias (right) by disease ($y_c = 2014$, sex = total, ages 20–89). For all diseases, MAPE grows steadily as the forecast horizon gets longer (Observation 1). The relative bias moves deeper into negative values for all diseases except **cancer** (Observation 2). (Reproduction: `reproduction/backtest/output/figures/overall_mape_bias_by_year.png`)

Figure 5.2: Heart disease (sex = total, three example ages): observed rates, Phase 1 smoothed rates, the Scale BB-D projection ($y_c = 2014$), and the actual rates 2015–2024. Scale BB-D (green dashed line) extrapolates the improvement trend of the training period as it is. So it stays systematically below the actual rates (red ×), which include the excess mortality of the COVID-19 period — a typical example of the negative bias in Observation 2. (Reproduction: `reproduction/backtest/output/figures/heart_disease_total_trajectory.png`)

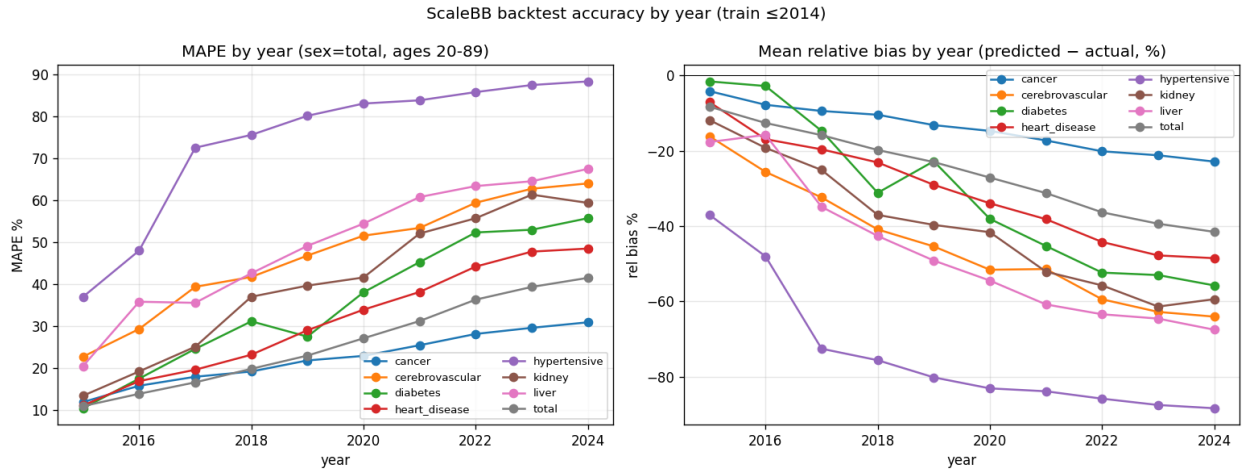


Figure 5: Yearly MAPE and relative bias over time (cutoff 2014, sex=total)

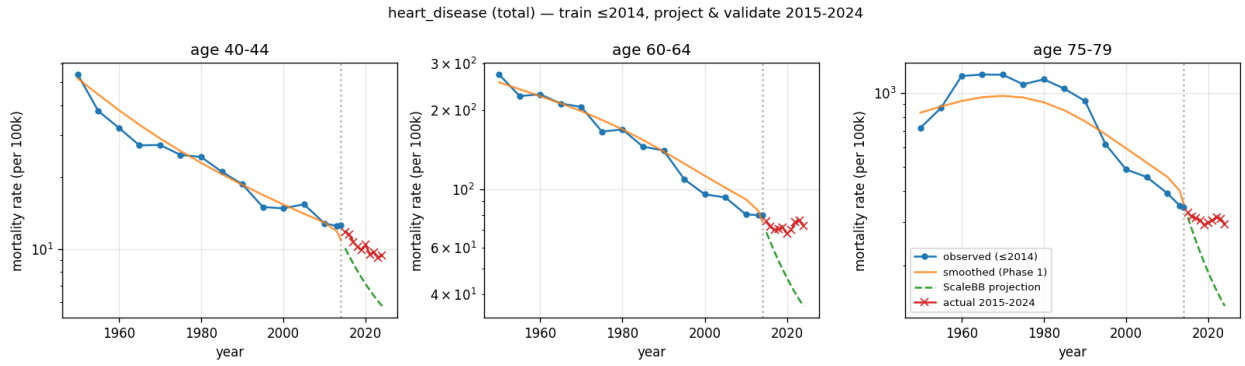


Figure 6: Rate trajectory of heart disease total — observed, smoothed, and Scale BB-D projection compared with actual 2015-2024

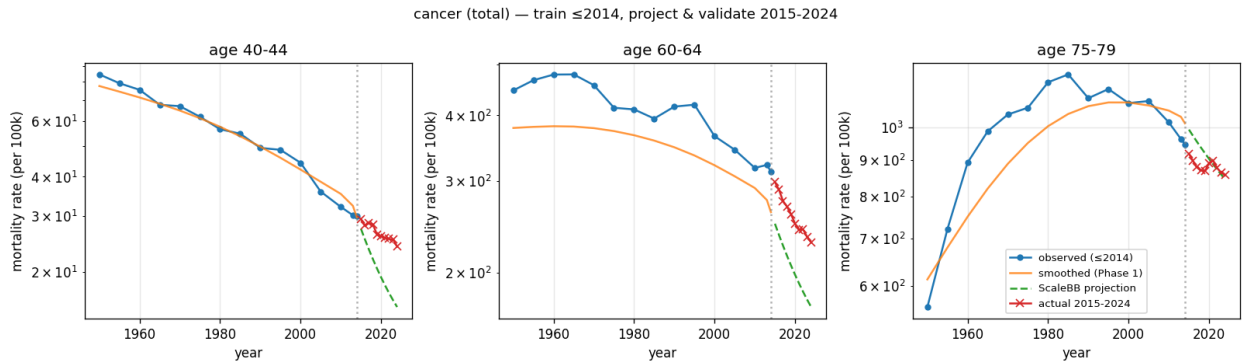


Figure 7: Rate trajectory of cancer (malignant neoplasms) total — observed, smoothed, and Scale BB-D projection compared with actual 2015-2024

Figure 5.3: The same trajectory for cancer (malignant neoplasms) (sex = total, three example ages). For **cancer**, the actual rates fall faster than the Scale BB-D projection, so the projection is above the actual rates (the positive bias in Observation 3). (Reproduction: `reproduction/backtest/output/figures/cancer_total_trajectory.png`)

5.2 Baseline comparison ($y_c = 2014$) — worse in all 24 cells

We compare Scale BB-D with the 3 baselines (§3.4.1) under the same settings.

Disease	scalebb	naive_last	mean_3pts	loglin_trend	Best baseline	Gap to best [pp]
cancer	22.41	12.78	14.92	5.86	loglin_trend	+16.55
total	26.01	8.73	12.79	5.74	loglin_trend	+20.27
heart_disease	31.27	14.91	18.29	8.28	loglin_trend	+22.99
diabetes	35.59	21.36	20.50	19.23	loglin_trend	+16.36
kidney	40.87	14.52	13.10	14.63	mean_3pts	+27.77
cerebrovascular	47.14	18.52	26.81	12.27	loglin_trend	+34.87
liver	49.49	10.44	10.17	17.51	mean_3pts	+39.32
hypertensive	73.83	39.29	30.33	35.88	mean_3pts	+43.50

Scale BB-D performs worse than all 3 baselines in all 24 cells (8 diseases $\times$ 3 sexes). Look at the gap to the best baseline. For diseases with a clear improvement trend (**cancer** / **total** / **heart_disease**), the gap to `loglin_trend` is +16 to +23pp. For diseases with an unclear trend (**liver** / **hypertensive**), the gap to `mean_3pts` is +39 to +44pp. Figure 5.4 shows the size of this gap. For sex = total, there is only one exception: **diabetes** in the single year 2015. Only there does Scale BB-D beat all baselines.

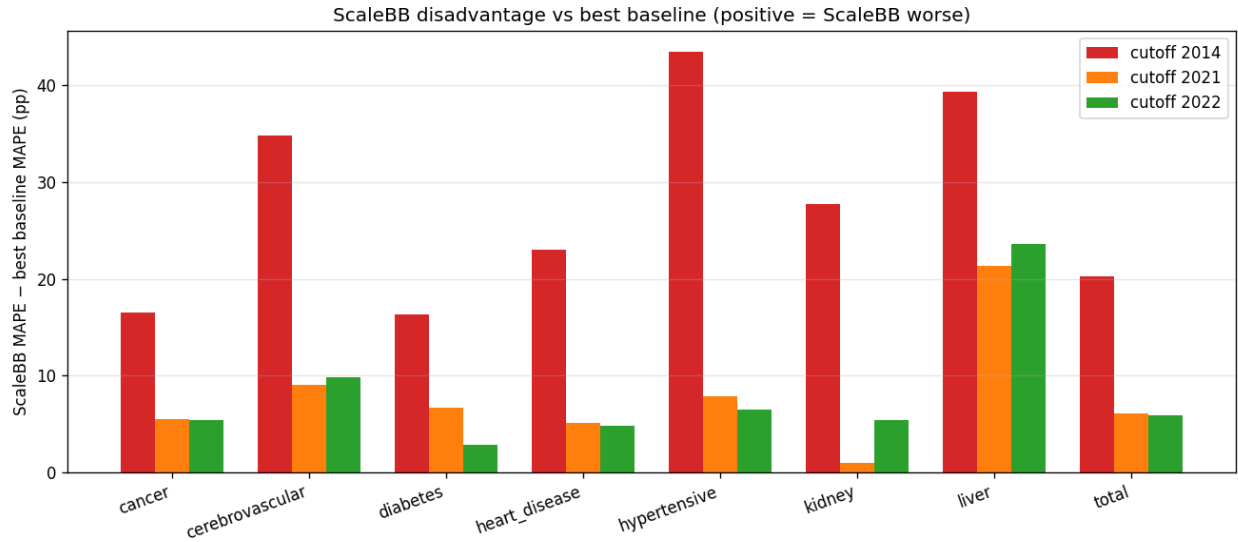


Figure 8: Gap between Scale BB-D MAPE and best baseline MAPE (by disease and by cutoff)

Figure 5.4: Gap of Scale BB-D MAPE — best baseline MAPE [pp] (positive = Scale BB-D performs worse, sex = total). At $y_c = 2014$ (red), Scale BB-D is worse by +16 to +44pp for all diseases. At $y_c = 2021/2022$ (orange / green), the gap shrinks, but the sign stays positive (§5.3). (Reproduction: `reproduction/backtest/output/cutoff_comparison/figures/scalebb_gap_vs_best_baseline.png`)

5.3 When the COVID-19 period is included in training ($y_c = 2021/2022$)

When we move the training cutoff to 2021 / 2022 and include the pandemic period in training, the MAPE of Scale BB-D shrinks dramatically.

Disease	$y_c=2014$	$y_c=2021$	$y_c=2022$
cancer	22.41	9.20	8.87
cerebrovascular	47.14	15.07	15.52
diabetes	35.59	16.02	13.29
heart_disease	31.27	10.27	8.37
hypertensive	73.83	24.13	20.53
kidney	40.87	16.69	15.59
liver	49.49	29.17	29.52
total	26.01	9.33	7.33

MAPE shrinks for all 8 diseases, in some cases by a factor of 6 to 8 (for example: **hypertensive** 73.83 $\rightarrow$ 24.13, **total** 26.01 $\rightarrow$ 9.33). At the same time, the absolute value of the bias also shrinks for all diseases. For **liver** at $y_c=2021$ ($-7.8 \rightarrow +0.1$ per 10^5) and for **hypertensive** at $y_c=2022$ ($-7.1 \rightarrow +0.4$), the sign turns from negative to positive. This shows the quantitative value of putting a structural break into the training data. As a measurement of the effect of COVID-19 on cause-specific mortality, it has meaning that is independent of the weak point forecasts (Figure 5.5).

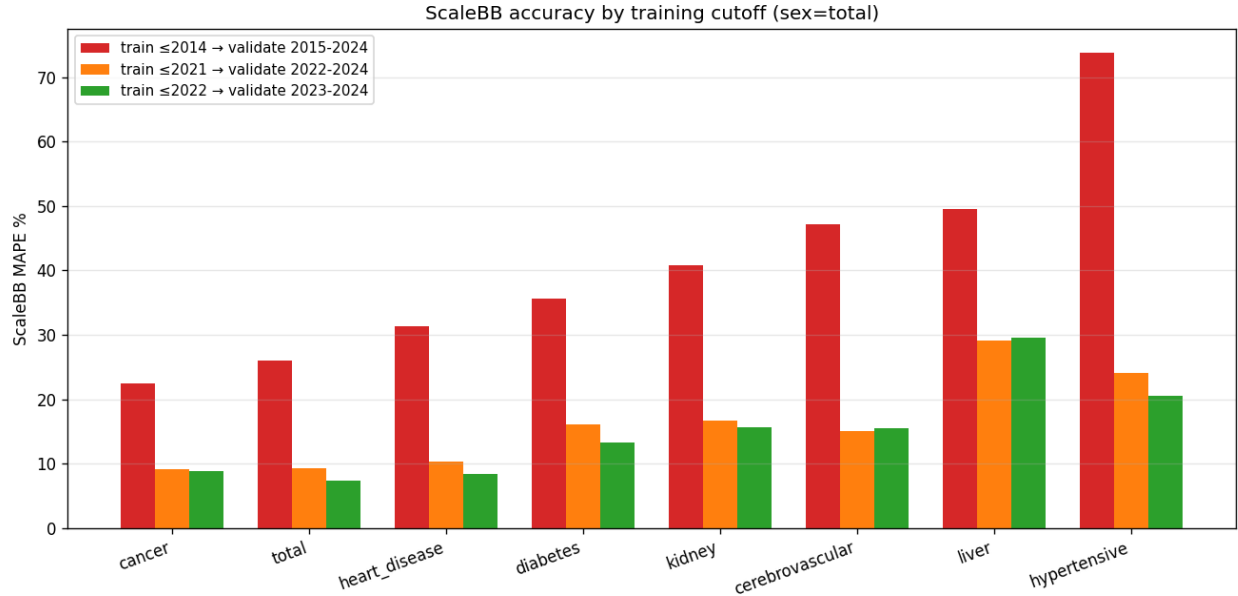


Figure 9: Scale BB-D MAPE by training cutoff (by disease)

Figure 5.5: Scale BB-D MAPE by training cutoff (sex = total). When the pandemic period is included in training — $y_c = 2021$ (orange) and $y_c = 2022$ (green) — MAPE shrinks strongly for all diseases compared with $y_c = 2014$ (red). Note, however, that the forecast horizon also becomes shorter at the same time (10 years vs 2–3 years) (§4.2). (Reproduction: [reproduction/backtest/output/cutoff_comparison/figures/scalebb_cutoff_comparison](#))

However, a smaller MAPE does not mean “catching up with the baselines.” The gap to the best baseline (Figure 5.4) shrinks from +16 to +44pp at $y_c=2014$ to +1 to +24pp at $y_c=2021/2022$. But for sex = total, Scale BB-D is still worse than the best baseline at every cutoff and for every disease. Count the cells where Scale BB-D beats “at least one baseline”: only 0 / 10 / 7 of the 24 cells at each cutoff. (It beats all baselines

in only 2 cells each at $y_c=2021/2022$.) With more training data, Scale BB-D reaches a level where it can compete with either `naive_last` or `loglin_trend`. For example, `kidney` at $y_c=2021$ beats `naive_last` (19.85) and `mean_3pts` (17.36), but it misses the best method, `loglin_trend` (15.69), by +1.00pp. It does not reach the best method.

5.4 Summary — a negative conclusion and the turn to the next chapter

The validation results on point forecast accuracy are clear.

- Short term (1–3 years ahead): the MAPE of Scale BB-D is 10–30%. This is an acceptable level in practice, but it is still worse than the baselines.
- Long term (10 years ahead): MAPE grows to 22–74%, and the gap is at its largest.
- At every cutoff and in all 24 cells, Scale BB-D is worse than the best baseline in MAPE.

If the original goal was “point forecast accuracy,” the conclusion of this chapter is a negative one: “Scale BB-D is not suitable for short-term point forecasts. `naive_last`, which carries the latest level forward, and the log-linear extrapolation `loglin_trend` are more accurate.” This result also fits the original design idea of Scale BB-D. It is a model that extracts long-term improvement trends over several decades. It is not a tool for hitting next year’s level with the smallest error.

But if we judge Scale BB-D on the single axis of point forecast accuracy, we may miss the essential output of this structural model — the improvement rates $i^*(x, y)$ with a guaranteed direction. In the next chapter, §6, we measure the directional accuracy DA (Equation 3.12), which the original goal did not measure. There the picture changes completely. The negative conclusion of this chapter turns from a “failure” into a “restatement of the question.”

The finding of this chapter — Scale BB-D is always worse than the baselines in point forecast MAPE — forms a pair with the directional accuracy finding of §6. The gap between the two is the core argument for repositioning Scale BB-D as a scenario generator in §7.

§6 Results II — The Directional Accuracy Finding

In this chapter we measure a second evaluation axis — directional accuracy DA (equations 3.11–3.12). The original validation plan (§4) did not measure it. We state the conclusion first. In §5, the point forecast MAPE fell behind most at $y_c = 2014$ (the 10-year-ahead forecast). At this same cutoff, the direction agreement rate for the six main diseases (all causes, cerebrovascular disease, heart disease, diabetes, cancer, kidney failure) reaches 77.7–95.0%. The same model lost every point forecast comparison. Yet it captured the sign of the trend (improvement or worsening) almost correctly. This gap is the central finding of this paper. All numbers below match the regenerated results of the reproduction package `reproduction/backtest/output/directional/` (sex = total).

6.1 Directional accuracy at $y_c = 2014$ — 77.7–95.0% for the six main diseases

We show the directional accuracy by disease for $y_c = 2014$ (forecasting the 10 years 2015–2024). This is the cutoff where the MAPE gap was largest. DA is evaluated only on cells where an actual change occurred (equation 3.12). Therefore we also report the number of valid cells n (§4.3).

Disease	n (valid cells)	DA [%]
total	140	95.00
cerebrovascular	134	91.04
heart_disease	137	90.51
diabetes	123	80.49

Disease	n (valid cells)	DA [%]
cancer	138	79.71
kidney	112	77.68
liver	118	27.97
hypertensive	110	23.64

When we place these results next to §5.1, the size of the gap stands out. For example, **cerebrovascular** had a MAPE of 47.14%. That was the third worst of the 8 diseases in point forecasts. Yet its direction agreement rate reaches 91.04%. **total** misses the 10-year-ahead level by 26% on average. Still, it gets the direction of change right in 133 of 140 cells. In other words, the error of Scale BB-D is not an error of “wrong direction”. It is an error of “going too far in the right direction” (over-extrapolating improvement). This is consistent with the dominance of negative bias observed in §5.1.

In contrast, two diseases miss the direction itself: **liver** (27.97%) and **hypertensive** (23.64%). Their mortality rates turned upward from the mid-2010s. However, the default setting ($L = +1\%$, continued improvement) kept producing projections in the improving direction. We treat this in detail in §6.5.

6.2 Across the 3 cutoffs — does the model capture direction regardless of the training period?

The table and Figure 6.1 show the Scale BB-D directional accuracy when we vary the training cutoff in three ways.

Disease	$y_c=2014$	$y_c=2021$	$y_c=2022$
total	95.00	42.86	60.71
cerebrovascular	91.04	69.23	51.85
heart_disease	90.51	50.00	52.00
diabetes	80.49	30.30	85.71
cancer	79.71	65.00	70.37
kidney	77.68	54.55	71.43
liver	27.97	45.95	52.00
hypertensive	23.64	68.97	85.00

Figure 6.1: Scale BB-D directional accuracy by training cutoff (sex = total). The dashed line is the chance level (a coin flip, 50%). At $y_c = 2014$ (red), the six diseases with continued improvement are far above the chance level, while **liver** / **hypertensive** are far below it. At $y_c = 2021/2022$ (orange, green) the pattern reverses, and **liver** / **hypertensive** recover (§6.5). (Reproduction: `reproduction/backtest/output/directional/figures/scalebb_directional_per_cutoff.png`)

Two points need care when reading Figure 6.1. First, $y_c = 2014$ is statistically the most stable. The number of valid cells is 110–140 at $y_c = 2014$. It is only 29–42 at $y_c = 2021$ and 20–28 at $y_c = 2022$ (§4.3). For cutoffs with a short validation period, the DA can move in steps of 10pp depending on the outcome of a few cells. Second, the lower values for short validation periods reflect the year-to-year swings just after COVID-19. For example, **total** at $y_c = 2021$ (42.86%) uses the excess mortality level of 2021 as the last observation. It then asks about the direction of the rebound in 2022–2024. This setting differs in nature from $y_c = 2014$, which asks about the sign of a trend over decades. The swing of **diabetes** from 30.30% ($y_c=2021$) to 85.71% ($y_c=2022$) shows the same short-term noise. The topic of this chapter is the direction of long-term trends. For that topic, the $y_c = 2014$ result — 77.7–95.0% for the diseases with continued improvement, with a 10-year validation period and 140 cells — is the most reliable estimate.

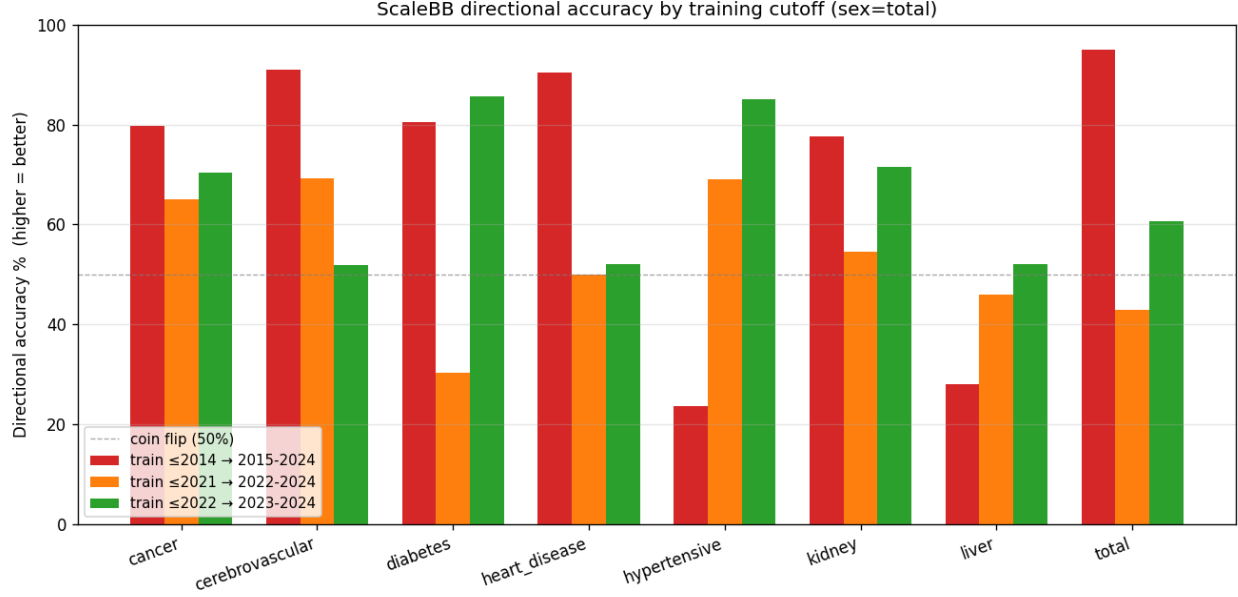


Figure 10: Bar chart of Scale BB-D directional accuracy by training cutoff and by disease

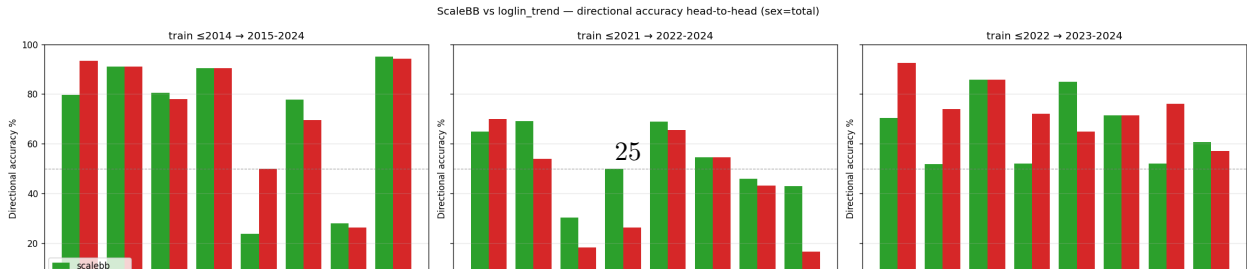
6.3 Baseline comparison — few methods carry direction information

We compare the DA of the four methods under the same definition. Over 24 comparisons (8 diseases $\times$ 3 cutoffs, sex = total), the win-loss record of Scale BB-D is as follows.

Compared against	Scale BB-D wins	Ties	Losses
vs <code>naive_last</code>	24	0	0
vs <code>mean_3pts</code>	15	0	9
vs <code>loglin_trend</code>	12	5	7

`naive_last` has $\Delta_{\text{pred}} \equiv 0$ by construction (equation 3.11). It carries no direction information at all (DA = 0%). In §5 this method was among the strongest on MAPE. On the direction axis, it cannot even enter the contest. This is the clearest sign of the gap between MAPE and DA. The direction information of `mean_3pts` is accidental. It predicts a “change” equal to the gap between the mean of the last 3 points and the observed value in the cutoff year. So it scores high when the cutoff-year value happened to dip: 76.36% for `hypertensive` and 61.02% for `liver`, both in a rising phase. But it misses almost systematically for diseases with clear improvement (4.29% for `total`). This is not direction information based on trend structure.

The only meaningful rival is `loglin_trend`. It holds explicit direction information through its OLS slope. Scale BB-D wins on net, with 12 wins, 5 ties, and 7 losses. But the details have structure. At $y_c = 2014$, the two methods are exactly equal for `cerebrovascular` (91.04%) and `heart_disease` (90.51%). For `total` they are nearly equal: 95.00% vs 94.29%. For diseases with a clear improvement trend, the long-term structural model and log-linear extrapolation reach the same sign. The losses of Scale BB-D concentrate mainly in `cancer` (it falls behind at all 3 cutoffs; 79.71% vs 93.48% at $y_c = 2014$) and in $y_c = 2022$, where the validation period is only 2 years. In contrast, at $y_c = 2021$ Scale BB-D is far ahead for `heart_disease` (50.00% vs 26.19%) and `total` (42.86% vs 16.67%). The 15-year OLS includes the COVID period, so the level shift pulls it off course and it misses the direction. Even in that phase, Scale BB-D sets the direction from decades of structure, and it degrades more gently (Figure 6.2).



extrapolator. It assumes that the log-linear slope of the last 15 years continues. Over short horizons (1–3 years), the level is almost fully set by the level at the end of observation and the recent trend. So these methods hold a structural advantage in point forecasts. Scale BB-D is designed differently. It respects the long-term trend extracted from more than 60 years of observations, and the convergence to the long-term rate L (equation 3.5). It gives these more weight than the local level near the end of observation. It deliberately ignores noise and short-term shocks near the end of observation. This is the root cause of its weaker short-term MAPE (§5.4).

What it gains in exchange is stability of direction. The sign of the improvement rate comes from structure over a span of decades. So it does not swing much, whatever phase the end of observation falls in. The robustness across cutoffs seen in §6.2 shows this. In other words, Scale BB-D is not a tool for hitting next year’s level with minimum error. It is a tool that extracts the direction and strength of the long-term trend explicitly, as an improvement rate $i^*(x, y)$ whose direction has been validated. As long as we ask about point forecasts of the level, the negative conclusion of §5 stands. But some uses ask about the sign of the improvement rate and the ability to control it — generating multiple scenarios. For those uses, this property becomes the requirement (§2.3, §7).

6.5 Handling direction-reversal diseases — the scope and limits of calibration

The direction misses of **liver** / **hypertensive** (DA 23.6–28.0%) are cases where the default setting of Scale BB-D, $L = +1\%$ (continued improvement), conflicted with the facts. These mortality rates turned upward from the mid-2010s. For practical use, the key question is whether this direction miss can be corrected. Here we separate and test two correction paths: (i) the calibration path — reset L and P per disease; (ii) the data path — include the recent post-reversal data in training. Table 6.1 and Figure 6.3 show the results.

Table 6.1: Recalibration experiment for direction-reversal diseases (sex = total, DA [%])

Setting	Path	liver	hypertensive
$y_c=2014$, default ($L=+1\%$, $P=2035$)	—	27.97 ($n=118$)	23.64 ($n=110$)
$y_c=2014$, $L=0\%$ (P unchanged)	Calibration	27.97	23.64
$y_c=2014$, $L=0\%$, $P=2020$	Calibration	33.05	30.91
$y_c=2021$, default	Data	45.95 ($n=37$)	68.97 ($n=29$)
$y_c=2022$, default	Data	52.00 ($n=25$)	85.00 ($n=20$)

Figure 6.3: Recovery of the direction agreement rate for direction-reversal diseases — the calibration path (reds: keep $y_c = 2014$ and reset L and P) and the data path (orange, green: include post-reversal data in training). The dashed line is 50%. (Generation script: `reproduction/backtest/make_calibration_recovery_figure.py`)

This experiment shows three facts.

1. Replacing the long-term rate alone does not change the direction within the validation period. Lowering L from $+1\%$ to 0% leaves the DA unchanged (the result is the same even down to -1%). The reason lies in the structure of equation (3.5). The whole validation interval 2015–2024 sits inside the blend interval $y_{\text{obs}} < y < P$ (§4.1). There, the improvement rate is a linear interpolation between the end-of-observation improvement rate $i(x, y_c)$ and L . Years closer to the end of observation put more weight on the $i(x, y_c)$ side. At $y_c = 2014$, $i(x, y_c)$ is positive, because it reflects the continued improvement of the training period. In addition, the end of observation falls on the boundary where the observation grid switches from 5-year steps to annual steps. So the smoothed value comes out larger than the real trend (the same boundary effect we noted in Figure 3.3). This anchor dominates the short-term projection direction, and L contributes almost nothing.
2. Moving the convergence year forward as well brings partial recovery, but it stays limited. We set $L = 0\%$ and also move P forward from 2035 to 2020. Then the dominance of the end-of-observation anchor ends

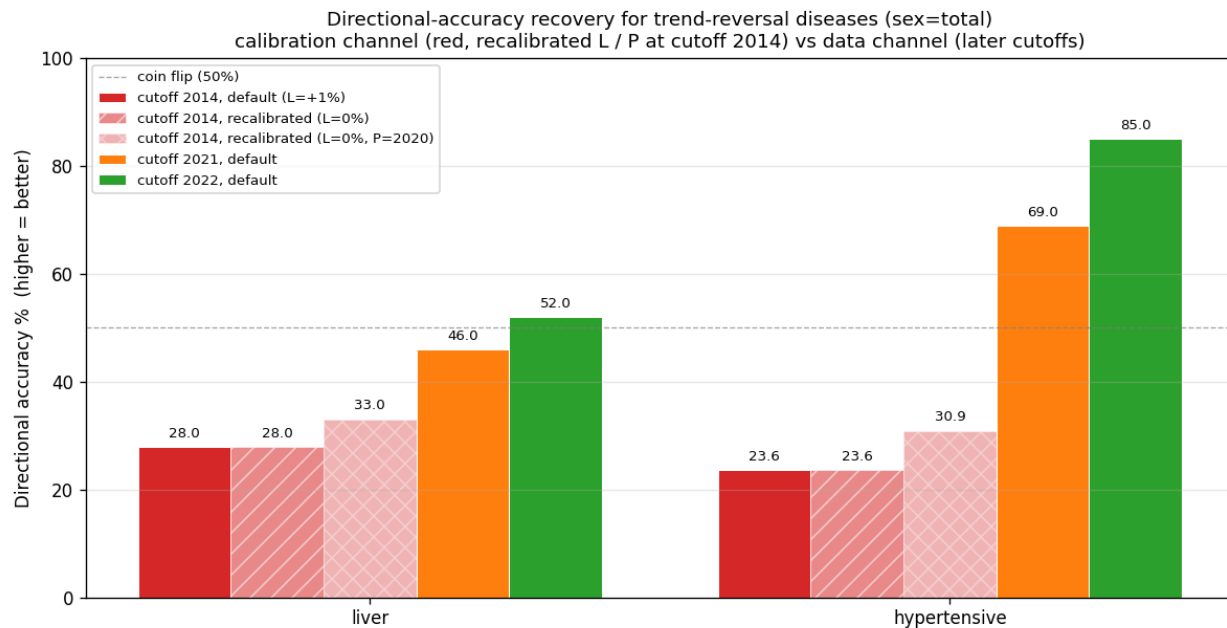


Figure 12: Direction agreement rate for direction-reversal diseases (liver / hypertensive) — comparison of the calibration path and the data path

earlier, and the DA improves to 33.05% / 30.91%. But it still stays below the chance level. Most of the validation period is still projected with the positive, pre-reversal improvement rate.

3. Real recovery happens through the data path. At $y_c = 2021/2022$, the training data include the post-reversal observations. With the default calibration left fully unchanged, **liver** recovers to 45.95 / 52.00% and **hypertensive** to 68.97 / 85.00%. The reason is that the sign of the observed improvement rate $i(x, y)$ extracted in Phase 1 is itself corrected by the data. Once the recent trend enters the training data, Scale BB-D corrects the direction structurally.

The practical implication of this separation is as follows. Disease-specific calibration of L and P controls the long-term direction after the convergence year. Its effect cannot be observed within a 10-year validation period. But over the projection horizon of insurance liability valuation (decades), that part becomes dominant. So resetting $L \leq 0$ for reversal diseases is a necessary condition for consistent long-term scenarios (see §9.1 for disease-specific calibration guidance). In contrast, the short-term direction near the end of observation is handled not by calibration but by regular refitting with the latest data. In other words, disease-specific long-term rate settings and regular refitting are not substitutes; they share the work. We add one more point. This correctability itself is a property unique to structural models. **naive_last** has no direction information to correct. Flipping the sign of the slope by hand in **loglin_trend** amounts to abandoning the premise of the estimation. Because the model holds the improvement rate and its long-term assumption as explicit parameters, we can identify a direction error, correct it, and validate it again.

Finally, we make clear what the limit of this verifiability means. The long-term effect of L itself cannot be verified directly until actual data after the convergence year accumulate (with the default setting, this needs actuals from 2035 onward). However, this is not a weakness specific to this framework. It is a general property of long-term improvement assumptions. Moreover, while we wait for actuals, the next structural break — a new epidemic or a shift in medical technology — can arrive. So the policy of “wait until enough data accumulate and then complete the validation” can never close in principle. The design of this framework goes the other way. The part that can be validated even with limited data — the direction of the improvement rate — becomes the foundation, fixed by the validation up to this chapter. The long-term assumptions that validation cannot reach are not embedded as implicit premises. Instead, we state them as a small number of

interpretable parameters (L , P) and turn them into objects to vary as scenarios (§7). Future structural breaks can also be handled. After the fact, regular refitting corrects the direction automatically (as the data path in this section demonstrated with COVID-19). Before the fact, we can quantify them as stress scenarios, such as $L \leq 0$ or catastrophe shocks. We do not try to hit long-term uncertainty with a point forecast. We handle the uncertainty explicitly, on top of a validated direction. This stance fits economic-value-based valuation (§2.3), which requires consistency across scenarios rather than a single point forecast.

The gap between “point forecasts lose everywhere” in §5 and “direction is right 77.7–95.0% of the time” in this chapter shows where the value of Scale BB-D lies. It is not in predicting single-year levels. It lies in the explicit output of the improvement rate $i^(x, y)$ whose direction has been validated. In the next chapter, §7, we discuss how this property translates one-to-one into the scenario generation structure (§2.3) required by economic-value-based valuation (ICS / IFRS 17).*

§7 Repositioning — Scale BB-D as a Scenario Generator for ICS / IFRS 17

In §5 and §6 we found a gap: the model is always worse in point forecast MAPE, but it predicts the direction of the improvement trend almost correctly. In this chapter we read this gap again through the two evaluation axes that we set out in §2.3. On the first axis (is the point forecast correct?), the negative conclusion of §5 does not change. However, economic-value-based evaluation (ICS / IFRS 17 / Solvency II, and in Japan the ESR) mainly asks for the second axis. That axis is: can the model generate a set of scenarios from one base in an internally consistent way? In this chapter we organize the meaning of the §6 findings for economic-value evaluation into three points (§7.1). We then discuss how the framework implements the scenarios that regulation requires (§7.2), the structural reasons why the baseline methods cannot play the same role (§7.3), the division of roles with the ESG (§7.4), the adoption burden (§7.5), and the direct application to death benefits contingent on specific diseases (§7.6).

7.1 Reading the findings again — three structural properties

For economic-value evaluation, we can organize the validation results up to §6 into the following three points.

1. Stable direction is the basic condition for consistency between scenarios. As we said in §2.3, the real substance of a regulatory calculation is “generating a set of scenarios that start from the same base”. The difference between base and stress must correspond one-to-one with the effect of the stress itself. If the sign of the trend that the model outputs is wrong, then the difference we get by moving the long-term rate L becomes a mixture of “the effect of the shock” and “calibration error”. We cannot use such a difference as a basis for capital calculation or disclosure. The backtest in §6 gives this basic condition — the sign of the improvement rate $i^*(x, y)$ is validated in advance — for the six main diseases (for direction-reversal diseases, we handle them by the division of roles in §6.5).
2. The explicit output of improvement rates makes direct implementation of shocks possible. As equation (3.6) shows, future rates are fully rebuilt from the base-year rates and the improvement rate path $i^*(x, y)$. Therefore every scenario definition can be written as an operation on i^* . Replacing a small number of interpretable parameters — the long-term rate L , the convergence year P , and the age taper $\tau(x)$ — is itself scenario generation. Methods that “do not have an improvement rate” do not have this operating surface (§7.3).
3. Two-dimensional smoothing guarantees a consistent age curve in all scenarios. The Phase 1 fit is shared between scenarios. Only Phase 2 (blending and projection) is run again for each scenario. So all scenarios start from the same smoothed age curve and the same base-year rates. The difference between scenarios comes only from the difference in the improvement rate paths. The four requirements listed in §2.3 (keeping the shape of the age curve, preserving relative relations, separating short term and long

term, and making many scenarios comparable with each other) are all satisfied at the same time by this structure.

7.2 Implementation mapping to regulatory scenarios

Table 7.1 maps the table in §2.3 (the rate-scenario inputs that are needed) to the implementation in this framework. The regulatory examples include the factors of the Pillar 1 regulatory notice (FSA Notice No. 74 of 2025 (Reiwa 7)), which we actually use in §8.

Table 7.1: Scenario types required by regulation and their implementation in this framework

Scenario type / calculation	Regulatory example	Implementation in this framework
Base scenario (current estimate, BEL)	Future rates $m(x, y)$ with assumptions re-evaluated at the valuation date	Use the projected rates of equation (3.6) as they are
Level shock	ESR: mortality +12.5% (notice Article 56), mortality −20% (longevity, Article 57), incidence rates for morbidity and disability +20% (Articles 59 and 60)	Uniform multiplication of the projected rates (multiply the Phase 2 output by a factor). ICS_C and ESR_M in §8
Trend shock	Change of the future improvement-rate trend (type classification in §2.3; the main areas are ORSA, internal management, and sensitivity disclosure, §8.5)	Replace $L \rightarrow$ only a Phase 2 re-projection. UP50 / DN50 / ICS_T in §8
Catastrophe shock	Infectious disease outbreak: mortality rises at once by 1/1,000 (notice Article 95)	Rewrite the first-year $i^*(x, y_0+1)$ so that $m(x, y_0+1)$ becomes +0.001, then re-project
Sensitivity disclosure (IFRS 17)	Change in liabilities when key insurance risk assumptions move within a reasonable range	Run a grid of $L \pm 50\text{bp}$, $P \pm 5$ years, and $a_0 \pm 5$ years of age
ORSA and medium/long-term stress	Quantify narrative scenarios such as worse public health or a stop in medical progress	Reset $L = 0$ or to a negative value (the behavior check for $L \leq 0$ is in §6.5)

The implementation column of Table 7.1 has one common point. All rows share the same Phase 1 fit. Each scenario is generated only by a Phase 2 re-projection with a changed `ScaleBBConfig` setting (or by multiplying the output by a factor). Phase 1 does not depend on L , so the marginal cost of adding a scenario is only one Phase 2 run. Also, with the APC extension (§3.3), cohort shocks — such as extra risk for a specific generation or faster improvement — can be implemented in the same pattern as operations on the cohort effect $\gamma(c)$.

The difference between a level shock and a trend shock is not just a difference in implementation. It appears as a difference in the age structure of the risk. To preview the numerical example of §8.4: the effect of the regulatory level shock (incidence rate +20%) on the present value of benefits is almost uniform across entry ages (+17 to 19%). The trend shock (zero improvement rate) has a clear age gradient: +31 to 32% for entry at age 30, and +10 to 11% for entry at age 60. The current method, which multiplies a static table by a uniform factor, cannot express a risk with this age structure in principle.

7.3 Why the baseline methods cannot play the same role

The baseline methods, which gave the most accurate point forecasts in §5, do not work as scenario generators for structural reasons.

`naive_last / mean_3pts` have no concept of an improvement rate. They can only express a multiplication of the level (a level shock). There is no place to define a trend shock — an intervention in the time structure of the rates. As we saw in §6.3, they also carry no direction information (or only by chance). So the very picture of a scenario as “a deviation from a base whose direction has been validated” cannot be built.

`loglin_trend` has an explicit trend, the OLS slope, and as §6.3 shows, its direction information is fairly good. However, it lacks the structure needed for scenario generation in three points. First, it has no time structure. It fixes the slope of the observation window used for estimation (the most recent 15 years) for the whole future. So it cannot express the Phase 2 time structure (equation 3.5) — “continue the observed trend in the short term, and converge to the assumed improvement rate in the long term” — or requirement 3 of §2.3, such as “a trend shock for 5 years only, then normal afterwards”. Second, it has no age structure. It is an independent OLS for each age, so it has no age $\times$ period interaction. It can shift the slope uniformly, but an age-dependent shock such as “improvement continues for young ages, and slows down for old ages” needs manual settings age by age. It also has no taper at high ages, so the extrapolation can easily diverge. Improvement rates can flip between neighboring ages, and the rate surface after a shock loses its smoothness. A rate curve that is inconsistent across ages cannot be used for product design or as a basis for capital calculation. Third, if we try to fix these problems from outside (manually reversing the slope, adjusting the slope age by age), we give up the assumptions of the estimation method. We fall back to ad-hoc table manipulation, and we lose the §2.3 property that “the difference corresponds one-to-one to the shock”.

From this we can say: the family of point forecasters and short-term extrapolators was built to “hit a single future”. It does not have the operating surface — an explicit output of improvement rates with validated direction — needed to “span several futures consistently”. Whether this operating surface exists or not is the essential difference that the MAPE comparison in §5 could not show.

7.4 Contrast with the ESG — a deterministic scenario generator for the disease risk axis

In insurance practice, the word “scenario generator” makes people think of the ESG (Economic Scenario Generator). So we make the difference in roles clear. The ESG is a device for the economic risk axes, such as interest rates and equities. It generates thousands of stochastic paths under a probability measure. It supports market-consistent valuation and the calculation of the time value of options and guarantees. In contrast, this framework covers deterministic scenario generation for the disease risk axis — for medical insurance, disease incidence rates (as a proxy); for death benefits contingent on specific diseases, cause-specific mortality (directly). The insurance risks in ICS / ESR require prescribed deterministic stresses, not stochastic paths (§2.3, §8.1). For the economic axes, a systematic generation device, the ESG, already exists. For the disease risk axis there is no counterpart. The current practice is to multiply a static assumed rate table by a uniform factor. This proposal fills this gap — assumption practice that cannot express a trend shock — and creates a new device that forms a pair with the ESG on the economic axes.

The two do not compete. They share roles and meet in the projection engine. The ESG supplies the discount rates and market scenarios. This framework supplies the rate tables for each scenario. The liability valuation model consumes both. The demonstration in §8 is calculated exactly in this setup — an ESR-compliant discount rate curve $\times$ claim rate tables by scenario. A stochastic extension based on the residual distribution of improvement rates (for example, bootstrap generation from a distribution of L) is a natural future extension of this framework, but it is outside the scope of this paper (§10).

7.5 A preview of the adoption burden

The phrase “creating a new generator” may suggest a large infrastructure investment. However, the burden of adding it to an existing workflow is low. The generator sits upstream of the projection engine. Its output is exactly the rate-table format that existing valuation practice already consumes. It works not as an addition to, but as a replacement of, the current process for making stress tables (static table $\times$ uniform factor). We discuss a systematic analysis of the adoption burden and a step-by-step adoption path in §9.2.

7.6 Direct application to death benefits contingent on specific diseases — two layers of applicability

Finally, we make the scope of this framework clear, following the double positioning in §3.1.3. For morbidity benefits in medical insurance, cause-specific mortality is a proxy for the incidence rate. The direction validation in §6 should be read as “validation of the methodology” — validation with incidence rates themselves is future work (§10). On the other hand, for products where cause-specific death is the benefit trigger — death benefits for the three major diseases, cancer death insurance, and disease-specific death riders by cause — the cause-specific mortality validated in this study is the assumption itself. We can apply the scenario generator directly, without the proxy caveat. Trend shocks through operations on L , and level shocks through the notice factors, feed directly into the sensitivity of the present value of benefits for these products. (In §8 we show that the same rate table, depending on how we read it, supports both regulatory cases: the morbidity stress (ESR_M) and the mortality stress (the level part of ICS_C).)

So the practical scope of this framework has two layers. For medical insurance (incidence-rate products), it works as a proxy application of a validated methodology. For death benefits contingent on specific diseases, it works as a direct application. In the second layer, the direction validation in §6 is itself the validation of that assumption. We do not claim to forecast the case fatality rate (mortality after diagnosis) in either layer (§10).

In this chapter, we read the §6 findings again as a repositioning: “the only method with an explicit output of improvement rates whose direction has been validated can generate the scenarios required by regulation, in a consistent way, just by replacing settings.” In the next chapter, §8, we put this generator on top of the ESR implementation details (discount rate curve and stress factors set by regulatory notice), and we answer the “So what” question with concrete numbers: BEL sensitivities.

§8 Financial Impact Demonstration

In this chapter, we show the practical meaning of the “repositioning as a scenario generator” from §7. We show it in numbers, as the sensitivity of the BEL (present value of benefits) in a simple projection model. The goal is not a precise valuation of liabilities. Instead, we show the direction and the size of the effect that changes in the long-term improvement rate L have on the present value of benefits. We present this in a form that can be applied directly in an economic-value-based evaluation framework. Japan started to apply the economic-value-based solvency regulation (ESR) on 31 March 2026. Within the limits of a simple model, we made the calculation follow the ESR implementation rules as closely as possible (§8.1).

8.1 Design of the Demonstration — A Simple Model Aligned with the ESR Implementation Rules

The calculation engine is a simple projection model. It is deterministic and works in annual steps. For a model point issued in 2026 with entry age x_0 , we value the present value of benefits as

$$\text{BEL}(x_0) = \sum_{t=0}^{T-1} P(t) S(t) q_{\text{dis}}(x_0 + t, 2026 + t) \text{SA} \quad (8.1)$$

$$S(0) = 1, \quad S(t+1) = S(t) (1 - q_{\text{dis}} - q_{\text{death}} - q_{\text{lapse}}) \quad (8.2)$$

Here q_{dis} is the sum of the scenario-specific claim rates for the 3 diseases (cancer, heart disease, and cerebrovascular disease). q_{death} is the projected all-cause mortality rate. It is fixed at the base scenario and shared by all scenarios, so that the sensitivity comes only from changes in the disease rates. q_{lapse} is the lapse rate. $P(t)$ is the discount factor for term t , and SA is the sum assured. We take the rate for each age and

each year from the cohort diagonal $m(x_0 + t, 2026 + t)$ of the rate surface $m(x, y)$ that the framework in §3 outputs.

The calculation follows the ESR implementation rules on the next 4 points.

1. Position of the BEL. Equation (8.1) corresponds to the “current estimate” part of insurance liabilities in the ESR. The current estimate is the present value of future cash flows, with the assumptions re-evaluated at the valuation date. In §8.5 we make a simple calculation of the aggregation of sub-risks within life insurance risk and of the MOCE (uncertainty margin, Margin Over Current Estimate). Aggregation across risk categories (with market, credit, operational risk, and so on), eligible capital, and the ESR ratio itself are outside the scope of this demonstration, because it has no asset side.
2. Discount rate. For the discount factor $P(t)$, we use a risk-free interest rate curve rebuilt by Smith-Wilson extrapolation. We use the parameters of the “yield curve creation tool” published by the Financial Services Agency (FSA), as of the end of March 2026. These are: the last observed maturity for yen interest rates, 30 years; the ultimate forward rate (UFR: Ultimate Forward Rate), 3.8%; the convergence maturity, 60 years; and the zero-coupon risk-free rates by maturity (13 observed maturities). The curve matches the observed maturities exactly. The instantaneous forward rate at the convergence maturity is within $\pm 1\text{bp}$ of the UFR. We do not apply spread adjustments such as the general bucket.
3. How we measure the sensitivity. In the ESR, life insurance risk is calculated by a “stress method”. This method measures the change in net assets when set stresses are applied to economic-value-based assets and liabilities. The sensitivity ΔBEL in this demonstration has the same form. If the asset side does not change, it corresponds directly to the fall in net assets under the stress method.
4. Use of the regulatory stress factors. For some scenarios (ESR_M and the level part of ICS_C in §8.2), we use the exact stress factors set by the Pillar 1 regulatory notice (FSA Regulatory Notice No. 74 of 2025). These are: incidence rate +20% for morbidity and disability risk (Articles 59 and 60: product class that pays a lump sum when a health event occurs, policy term over 5 years, Japan), and mortality rate +12.5% for mortality risk (Article 56: Japan).

At the same time, we state the limits of the simple model clearly. The cash flows include benefits only. Premiums and expenses are not included (the valuation is a present value of benefits). We treat decrements with an independence approximation. The all-cause mortality rate includes deaths caused by the 3 diseases. This effect on the sensitivity is second-order, because the rate is shared by all scenarios. Also, as stated in §3.1.3, the rates are cause-specific mortality rates. They are a proxy for the incidence rates of medical insurance. Re-running the demonstration on a practical projection model that insurance companies already use is a task for future validation (§10).

8.2 Scenarios and Model Points

There are 6 scenarios, listed below. All of them share the same fit (Phase 1) from §3.2. UP50/DN50/ICS_T only replace the setting L and re-run the Phase 2 projection. ICS_C and ESR_M are generated only by a uniform multiplication of the rates.

SCN_CD	Content	How it is generated	Regulatory meaning
BASE	Base ($L = 1.0\%$)	Default settings	Base scenario for the current estimate
UP50	Faster improvement ($L = 1.5\%$)	Replace L	IFRS 17 sensitivity disclosure +50bp
DN50	Slower improvement ($L = 0.5\%$)	Replace L	IFRS 17 sensitivity disclosure −50bp
ICS_T	Improvement stops ($L = 0\%$)	Replace L	Trend shock (zero improvement rate)

SCN_CD	Content	How it is generated	Regulatory meaning
ICS_C	Improvement stops + level	$ICS_T \times 1.125$	Combination of trend and mortality level shocks (+12.5% is the same level as the mortality risk factor in Article 56 of the notice)
ESR_M	Morbidity stress	$BASE \times 1.20$	Articles 59 and 60 of the notice, morbidity and disability risk (lump-sum class, long term, Japan: incidence rate +20%)

The model points cover a product that pays a lump sum of 1 million yen for the three major diseases (at first diagnosis), with coverage to age 90. We place it at 8 points: entry ages 30/40/50/60 $\times$ male and female (MP01–MP04 male, MP05–MP08 female). The lapse rate is fixed at 3% per year. Death exits follow the projected all-cause rate. Both are shared by all scenarios.

The scenario table matches the two layers of applicability in §3.1.3 in the following way. The same rate table can be read in two ways. (i) If we read it as a proxy for the disease incidence rates of medical insurance, it falls under morbidity and disability risk (ESR_M). (ii) If we read it as an assumption for death benefits contingent on specific diseases, it falls under mortality risk (+12.5%, the same level as the level part of ICS_C). In other words, the single table of this demonstration maps to a regulatory stress under either of the two readings.

8.3 Results — One Table of BEL Sensitivities

Table 8.1 shows, for each model point, the BASE-scenario BEL (in yen, per 1 million yen of sum assured) and the change rate of each scenario compared with BASE.

Table 8.1: BEL sensitivity by model point (change rate vs BASE, %)

MP	Entry age and sex	BASE BEL (yen)	UP50	DN50	ICS_T	ICS_C	ESR_M
MP01	30 M	14,420	−12.6	+14.5	+31.0	+45.3	+18.3
MP02	40 M	29,094	−9.5	+10.5	+22.2	+35.5	+18.1
MP03	50 M	56,215	−6.7	+7.3	+15.1	+27.5	+17.7
MP04	60 M	100,454	−4.4	+4.6	+9.5	+21.1	+17.2
MP05	30 F	9,681	−12.5	+14.7	+31.8	+47.1	+19.1
MP06	40 F	19,221	−9.7	+10.9	+23.3	+37.6	+18.9
MP07	50 F	36,139	−7.2	+7.9	+16.5	+29.9	+18.7
MP08	60 F	62,764	−5.0	+5.3	+11.0	+23.7	+18.4
Total	—	327,988	−6.6	+7.2	+15.0	+27.7	+18.0

Figure 8.1: BEL sensitivity by model point and scenario (change rate vs BASE, %). The horizontal axis shows the model points (entry age and sex, ordered by entry age). The series are the 5 scenarios UP50 / DN50 / ICS_T / ICS_C / ESR_M. (Generation script: `ScaleBB/Research/scripts/bel_demo/aggregate_bel_results.py`)

We can summarize the results in 3 points. First, the order $UP50 < BASE < DN50 < ICS_T < ICS_C$ holds at every model point. The results are monotonic in L , so changes in L translate into the present value of benefits in a consistent way. Second, the sensitivity has a clear age gradient. It is larger when the entry age is younger (under ICS_T, +31 to +32% at age 30 against +10 to +11% at age 60). A change in the improvement trend builds up like compound interest over the remaining term. So longer contracts feel a larger

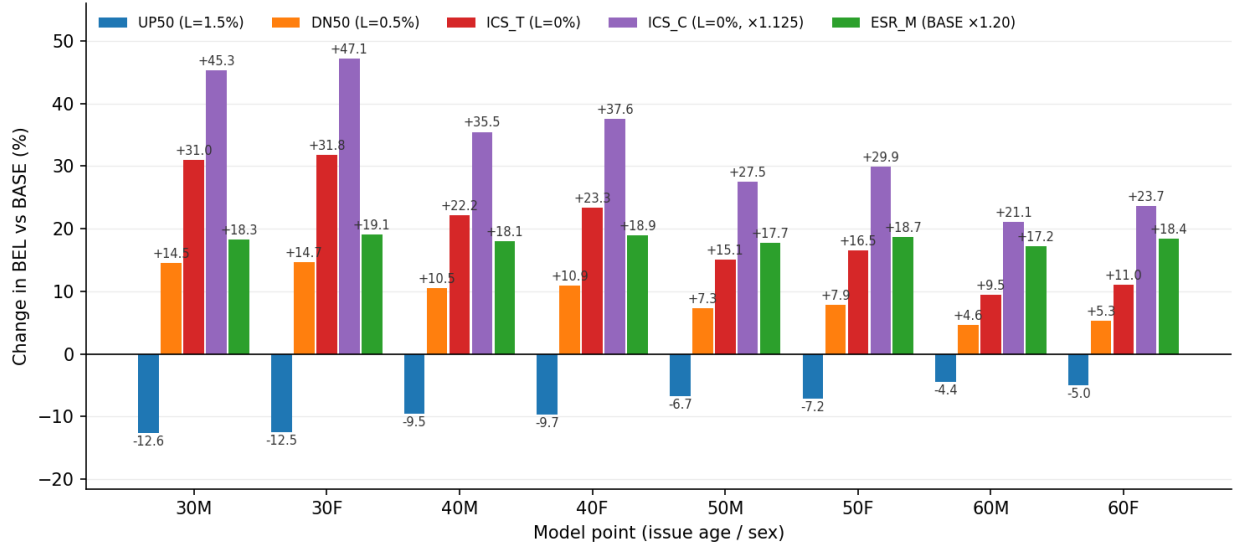


Figure 13: BEL sensitivity by model point and scenario (change rate vs BASE, %, ordered by entry age)

effect, which matches intuition. Third, for women the BASE level of the present value of benefits is lower than for men. However, their sensitivity in change-rate terms is slightly higher than men's in all scenarios.

8.4 Discussion — Trend Shock versus Level Shock

The most telling point in Table 8.1 is the contrast between ESR_M and ICS_T. The effect of morbidity and disability risk under the stress method of the notice (ESR_M: a uniform level shock of +20% on incidence rates) is almost constant across entry ages (+17.2 to +19.1%). This is a natural result of the structure of current practice (§2.3), which multiplies a static rate table by a uniform factor. In contrast, the effect of the trend shock (ICS_T: zero improvement rate) has an age gradient from +9.5% to +31.8%. In total, it is close to the regulatory level shock (+15.0% versus +18.0%). But at age 30 it is about 1.7 times the regulatory level shock, and at age 60 it is a little under 60% of it.

This contrast has 2 practical implications. First, the uncertainty of the improvement trend concentrates in young, long-term contracts. If we capture risk only with a uniform level factor, we may systematically under-estimate (or over-estimate) trend risk, depending on the age mix of the portfolio. Second, a shock with this time structure can only be generated by a framework that outputs the improvement rate $i^*(x, y)$ explicitly (the limit of the baseline methods in §7.3). In our framework, we can generate it just by replacing L . Moreover, as §6 showed, the backtest has validated its direction in advance. The standard method in Pillar 1 of the regulation is based on level shocks. Our framework can therefore be applied as it is, with no extra changes, to run trend scenarios as a complement in ORSA, sensitivity disclosure, and internal management (Pillar 2).

Also, the level part of ICS_C ($\times 1.125$) is the same level as the mortality risk factor in Article 56 of the notice (Japan +12.5%). So for death benefits contingent on specific diseases (Novelty B, the layer of direct application), ICS_C keeps a direct regulatory reading as “a combination of a trend shock and the regulatory mortality stress”. For medical insurance (the layer of proxy application to incidence rates), ESR_M is the matching scenario. Under either of the two readings, the sensitivities of this demonstration can be interpreted in the vocabulary of regulatory stresses.

We discuss the implementation burden in §9.2.

8.5 Extension — Simple Calculation of Life Insurance Risk Capital and Insurance Liabilities

The sensitivities up to the previous section were Δ BEL for individual stresses. In this section, we go one step further into the standard method of the Pillar 1 notice. We calculate the 5 sub-risks of life insurance risk for this portfolio, using the stress factors of the notice (all with the geographic class Japan). We then continue to correlation aggregation (Article 81) and the MOCE (Article 29), and show the results.

Table 8.2: Simple calculation of life insurance risk capital and insurance liabilities (total of the 8 model points, yen)

Item	Amount (yen)	Applied provision of the notice
Sub-risk: mortality	0	Article 56 (mortality rate +12.5%. In this product, death works as an exit, so it does not reduce net assets, and the amount is 0)
Sub-risk: longevity	11,698	Article 57 (mortality rate $-20\% \rightarrow$ more survivors, so the present value of benefits rises)
Sub-risk: morbidity and disability	58,917	Articles 59 and 60 (incidence rate +20%. By definition this equals the Δ BEL of ESR_M in Table 8.1)
Sub-risk: lapse	50,096	Articles 61 to 63 (the larger of a level-and-trend shock of $\pm 25\%$ and a mass lapse of 30%. For this product, the lapse-decrease side binds)
Sub-risk: expense	0	Article 64 (this demonstration has no expense cash flows)
Life insurance risk (after correlation aggregation)	80,067	Correlation matrix of Article 81 (diversification effect of -34% against the simple sum of 120,711)
Current estimate (total BASE BEL)	327,988	Equation (8.1)
MOCE	32,286	Articles 29 and 30 (cost-of-capital rate 3%, approximation of the run-off pattern of the estimated required capital. 9.8% of the current estimate)
Insurance liabilities (current estimate + MOCE)	360,273	—

This calculation has 3 implications. First, the largest component of life insurance risk for this product (morbidity benefits) is morbidity and disability risk. Its amount comes directly from the ESR_M scenario in Table 8.1. In other words, the rate-table output of our framework can be applied in the same form, not only to sensitivity analysis but also to the stress-method calculation of required capital. Second, mortality risk is 0 because, in this product, death works on the exit side (it reduces benefits). If we read the same rate table as death benefits contingent on specific diseases (Novelty B), the mortality stress of +12.5% applies directly to the claim rates (the level part of ICS_C). This contrast shows clearly the regulatory difference between the two layers of application. Third, the standard method of the regulation, up to this point, consists

only of static level stresses. The uncertainty of the improvement trend (the risk with an age gradient, shown in §8.4) does not appear in the standard method of Pillar 1. The quantitative capture of trend scenarios belongs to ORSA and internal management (Pillar 2) and to sensitivity disclosure (Pillar 3). That is where our framework will mainly be implemented (§9).

Note that the calculation in this section is an illustration. It applies the standard method to a simple portfolio with a single product and benefit cash flows only. It does not include the setting of homogeneous risk groups, expense and premium cash flows, or aggregation across risk categories, which a real required-capital calculation needs.

The demonstration in this chapter showed one thing. The explicit output of improvement rates, whose direction was validated in advance (§6–§7), translates into concrete sensitivity numbers on the implementation rules of the first year of ESR application. In the next chapter, §9, we discuss the calibration guidance and the implementation burden for introducing this into practice.

§9 Practical Implementation Guidelines

In §7 we repositioned this framework as a scenario generator. In §8 we quantified its sensitivity. This chapter gives guidance for bringing the framework into the assumption workflow of practical work. §9.1 gives guidance for disease-specific calibration based on the backtest results (§5–§6), and for tuning the hyperparameters (§3.2.3). §9.2 gives a systematic answer to an expected objection: “Building a new generator may be a heavy burden to introduce.” It also shows a step-by-step introduction path. §9.3 reports a real case of integration into an existing system.

9.1 Guidance for Calibration and Tuning

Premise: two channels share the work. As we separated in §6.5, disease-specific calibration (resetting L and P) controls the long-term direction after the convergence year. The short-term direction near the end of the observed data is handled by regular refitting with the latest data (the data channel). Therefore, this section gives guidance on two things together: “disease-specific settings of L and P ” and “the operation of regular refitting”. The first alone cannot fix short-term direction errors (the experiment in §6.5). The second alone does not secure consistency after the convergence year — the period that dominates the projection horizon of insurance liabilities.

Disease-specific long-term rate L . Table 9.1 shows guidance by disease group, derived from the backtest results (the default is $L = 1\%$ for all diseases, §3.2.3).

Table 9.1: Calibration guidance for the long-term rate L by disease group

Disease group	Target diseases	Guidance for L	Supporting validation results
Clear improvement trend	cancer / heart_disease / cerebrovascular / total	Keep the default 1%, or consider raising it to 2%	DA 79.7–95.0% ($y_c=2014$, §6.1). cancer shows under-extrapolation of improvement (positive bias, §5.1), so there is room to raise L

Disease group	Target diseases	Guidance for L	Supporting validation results
Watch for a turn to worsening	diabetes / kidney	Set a modest value of 0.5 to 1%	DA is high (77.7–80.5% at $y_c=2014$), but the direction is unstable when the observed data ends around the COVID period (§6.2)
Direction reversal	liver / hypertensive	0 or below (0 to -1%)	With the default settings, the model misses the direction itself (DA 23.6–28.0%, §6.5). Resetting to $L \leq 0$ is a necessary condition for long-term scenario consistency

When applying Table 9.1, one must note the period in which the effect of resetting L appears. Setting $L \leq 0$ for reversal diseases changes the direction agreement rate very little within the backtest validation period (10 years) (§6.5, Table 6.1). This is not a flaw in the guidance. It happens because the validation period lies inside the blending interval, where the anchor at the end of the observed data dominates. It reflects the division of work: the refitting described below handles the short-term direction within the validation period, and L handles the long-term direction after the convergence year.

Convergence year P . The default is 2035. In some situations, the improvement rate at the end of the observed data differs greatly from the long-term assumption — right after a structural break, or for direction-reversal diseases. In those cases, consider moving P earlier, to 2025–2030. This releases the dominance of the end-of-data anchor sooner and speeds up the return to the long-term assumption. (In §6.5, DA within the validation period recovered partly only when $L = 0\%$ was combined with an earlier P .) In contrast, when the continuation of the recent trend matters most, the default can stay as it is.

Smoothing strength λ_{year} (`lam_col`). The default is 40 (§3.2.3). Use different values for different purposes. For monitoring and analysis, where you want the model to follow structural change, lower it to 10–20 so that it respects the local trend at the end of the observed data. For uses where year-to-year stability of the rates matters, such as generating forward-looking assumed rate tables, raise it to 60 (the age20 preset in the note of §3.2.3). The age-direction λ_{age} and the difference order d were robust at their defaults (40 and 2) within the scope of this validation.

Regular refitting. As a basic rule, refit once a year, in line with the annual release of the Vital Statistics of Japan (MHLW). As the data channel in §6.5 shows, short-term self-correction for direction reversals and structural breaks works through refitting, not through calibration operations. (This was demonstrated by the reflection of COVID-19; §5.3 and §6.2.) The computing load of refitting is small (§9.2, argument 4), so it is not an obstacle to annual operation.

A pattern for model validation. When you extend the model to new diseases, new granularity, or new products, we recommend measuring DA before applying it. Use the 3-cutoff backtest of this paper (§4) and the directional accuracy (equation 3.12) as the template. The validation procedures of §5–§6 can be reused directly as a model-validation template at each company.

9.2 Analysis of the Introduction Burden — Five Arguments and a Step-by-Step Path

The proposal to “build a new scenario generator for the disease-risk axis” (§7.4) may bring to mind a large system investment and operational change, as the introduction of an Economic Scenario Generator (ESG) once did. However, the introduction burden of this framework is low, for the following five reasons.

1. Upstream position, same output format. The generator sits upstream of the projection engine. Its output is a format that existing valuation practice already consumes: rate tables by age x calendar year (or age x policy duration). Downstream steps — the valuation model, the closing process, and reports — need no change. In fact, the demonstration in §8 reached a sensitivity table using only the existing rate panel and the algorithm core. The only thing it asked from the valuation side was to read scenario-specific rate tables.
2. Replacement, not addition. Current practice already has a step that builds rate tables for stress (static table x uniform factor). This framework replaces that step. It does not increase the number of scenario runs. A level shock that the uniform-factor method could express can still be expressed as a uniform multiplication in this framework (Table 7.1). The output of the current method can be reproduced as a special case of this framework.
3. No new data preparation is needed. The inputs are public statistics (the Vital Statistics of Japan) and the company’s own experience data, which each company already collects for experience-rate analysis. There is no need to buy new external data or to build a new data-collection system.
4. The computing load is light. The framework is deterministic, so it does not need the infrastructure for computing and storing thousands of paths, as an ESG does. On the generator side (creating rate tables), the Phase 1 fit is shared across all scenarios. The marginal cost of adding a scenario is only one Phase 2 re-projection (§7.2). On the valuation side (computing the present value of benefits), the only requirement is reading scenario-specific rate tables (argument 1). Since these tables have the same format as the existing ones (arguments 1 and 2), the extra load on the existing valuation batch stays linear in the number of runs. Measuring the actual run load and the integration effort on a practical projection model is left to future validation (§10.4).
5. The governance burden is low. The judgment parameters are essentially only L and P . Both are quantities that can be explained to management and validation teams: “how much improvement do we assume will continue in the long term?” and “when do we converge to the long-term assumption?”. The computation is deterministic and reproducible. For model validation, the backtest of this paper (§4–§6) serves directly as the template (§9.1). This differs in kind from black-box machine-learning models, where explainability itself becomes the issue for validation and explanation.

A step-by-step introduction path. Introduction does not require a full switch at once. A design is possible that promotes the framework step by step, starting from low-risk uses.

- Stage 1 — Trial use in internal sensitivity analysis and ORSA (Own Risk and Solvency Assessment): Outside regulatory calculations, run the framework in parallel with the current uniform-factor method. Bring the extra information from trend scenarios (the age gradient in §8.4) into internal management and risk understanding. This can start without changing current practice at all.
- Stage 2 — IFRS 17 sensitivity disclosure: Use grid runs such as $L \pm 50\text{bp}$ (Table 7.1) for the calculation of disclosed sensitivities.
- Stage 3 — Regulatory calculations under the Insurance Capital Standard (ICS) and economic-value-based solvency regulation (ESR): Extend the scope to required-capital calculations under the stress method (§8.5).

Whether to promote the framework at each stage can be judged on evidence, by passing the backtest validation of §9.1 (measuring DA on the target of application). A transition that starts with parallel operation and comparison with the current method acts as a buffer. It does not demand a “jump away from uniform factors”.

9.3 A Real Case of Integration into an Existing System

The algorithm core of this framework is a small implementation. It consists of pure smoothing and projection functions (`fit_scale_bb` / `project_scale_bb`) and a configuration object (`ScaleBBConfig`) (§3.2). This core has already been integrated into the authors’ experience-rate aggregation process system. Within one workflow, it can run from the aggregation of the company’s own experience data to the generation of projected rate tables using disease-specific presets. The integration required only copying the core and adding configuration presets. No changes to the existing aggregation and output steps were needed. In this form, a

projection function is added downstream of the existing experience-rate analysis steps. This case supports, at the implementation level, argument 1 of §9.2 (upstream position, same output format) and argument 3 (existing data is enough).

In this chapter, Scale BB-D as a scenario generator became concrete, down to “how to configure it and how to introduce it”. The next chapter, §10, faces the limitations of this study — the proxy nature of the data, the limits of long-term verifiability, and the simplicity of the financial demonstration — and shows a path toward future work.

§10 Limitations and Future Work

In this chapter, we organize the limitations of this study into 5 areas. For each area, we describe the related future work. We already mentioned most of these limitations in earlier chapters. Here, we place them in the context of the whole paper. We also give concrete paths for future validation and extension.

10.1 Proxy nature of the data — Limited scope of application and direct validation with incidence rates

The first limitation is the proxy nature of the input data. This study validated cause-specific mortality rates. For the disease incidence rates of medical insurance (benefits paid on diagnosis), our work is only a proxy validation of the methodology (§3.1.3). Between incidence rates and cause-specific mortality rates, there is a third factor: the case fatality rate (mortality after diagnosis), which changes over time. Because of this, our validation cannot separate differences in level and trend. So the application to incidence-rate products is indirect. It rests on this basis: the methodology was validated on series that share the same age and cohort structure. However, this limitation applies only to incidence-rate products. For death benefits contingent on specific diseases (Novelty B), cause-specific mortality is the assumption itself. So the proxy problem does not appear there (§7.6).

The future task is direct validation with incidence rates. To build incidence rates themselves, the best path uses databases that follow individuals over time. Examples are health insurance society claims data (medical billing records) and health check-up records. With the insured group as the denominator and first disease events by disease as the numerator, we can build incidence rates by year and age. These rates do not depend on medical treatment rates or mortality rates. Then a backtest of the same form as in this paper becomes possible for all diseases. However, these databases are sold as paid big data. At present, the cost is a large barrier for research use.

We also confirm again that the case fatality rate (mortality after diagnosis) is outside the scope of this study. The case fatality rate needs the number of diagnosed cases as its denominator. We cannot get it from cause-specific mortality per total population. Estimating the case fatality trend as cause-specific mortality divided by observed incidence is future work. This study does not claim, in any form, that it can predict the prognosis after diagnosis.

10.2 Testability of long-term forecast accuracy — Handle it through scenarios, not by waiting for data

The second limitation is that the long-term effect of the long-term rate L itself is not yet validated. As we said in §6.5, L and P control the direction after the convergence year. Under the default setting ($P = 2035$), no actual data exists yet for that period. The 10-year backtest validated the short- to mid-term direction in the blending interval.

However, this study does not treat this limitation in the frame of “wait for data to build up, then complete the validation”. The reasons and the response follow the structure discussed at the end of §6.5. (i) The part

we can validate with limited data — the direction of improvement rates — was fixed as the foundation by the validation in this paper (§6). (ii) The long-term assumptions that validation cannot reach are not hidden inside the model as implicit premises. We made them explicit as the interpretable parameters L and P . We turned them into objects that we vary as scenarios (§7–§8). (iii) While we wait for actual data to build up, the next structural break — a new infectious disease outbreak, or a shift in medical technology — can enter the data. So a validation that waits for data can never finish, as a matter of principle. Against structural breaks, two defenses work. After the event, regular refitting corrects the direction by itself (the data path in §6.5, already shown with COVID-19). Before the event, we can quantify the risk as stress scenarios, such as $L \leq 0$ or a catastrophe shock (Table 7.1). We do not try to remove long-term uncertainty by hitting point forecasts. Instead, we handle uncertainty explicitly on top of the validated direction. This stance fits the requirements of economic-value-based evaluation (§2.3). We believe it is a realistic way to support the reliability of disease-rate assumptions when only limited data is available.

10.3 Building a systematic method for disease-specific calibration

The third limitation concerns the disease-specific calibration guidance in §9.1. It is provisional. We induced it from a backtest of 8 diseases $\times$ 3 cutoffs. Future work has the following 3 points. First, upgrade the choice of L and P to a systematic selection procedure. It should be based on directional accuracy in pseudo out-of-sample tests with shifted cutoffs. Second, check that the guidance stays stable at finer disease levels, such as ICD mid-level categories, and at finer age levels than 5-year age groups. Third, validate projections that include the cohort effect of the APC extension (§3.3), using the same backtest design (§4). This would give validation support to the practical use of cohort shocks (§7.2).

10.4 Simplicity of the financial demonstration and re-running it on a practical model

The fourth limitation is about the BEL sensitivity demonstration in §8. It is a demonstration with a simple projection model that uses benefit cash flows only and an independence approximation for lapses, and with 8 model points. For financial validation at a practical level, the following work remains. First, re-run it on a practical projection model that insurance companies already use in production. Then reconcile the results at the rate level and the present-value level, including measuring the actual run-time load. Second, validate the financial impact on a real in-force portfolio. This includes the policy distribution, premiums, expenses, and lapse behavior. Third, add the parts of the ESR implementation we did not apply. These are: integration across risk categories, eligible capital and the ESR ratio, the spread adjustment (the general bucket and similar items), and a more precise treatment of MOCE and required-capital integration, which §8.5 only approximated.

10.5 Stochastic extension

Finally, this framework is designed as a deterministic scenario generator. That is enough for the deterministic stresses that the standard regulatory method requires (§2.3). But a stochastic view helps with understanding the risk distribution in ORSA and with quantifying risk appetite. A natural direction is to add a stochastic layer on top of the validated deterministic core. One example is bootstrap generation of L based on the distribution of improvement-rate residuals (§7.4). Integration with an Economic Scenario Generator (ESG) lies on the same line. This means building a systematic operation where disease-risk scenarios and economic-risk scenarios feed the same projection.

None of these limitations overturns the central conclusion of this study — that explicit improvement-rate outputs with a validated direction work as a scenario generator. Instead, they define its scope of application and the path to refine it to a practical level. The next chapter states the conclusions of this paper.

§11 Conclusion

This paper extended Scale BB, the mortality improvement scale of the SOA (2012), from all-cause mortality to cause-specific (disease-specific) mortality. We tested it with a 3-cutoff backtest on a panel from the Vital Statistics of Japan (MHLW) for 1950–2024. The results form three steps. First, the original goal failed: the model is not useful for point forecasts. Scale BB-D performs worse in MAPE than the best simple baseline in all cutoffs and all 24 cells (+16 to +44pp at $y_c = 2014$, §5). Second, we measured the same output again by the direction of change. The directional accuracy for the 6 main diseases reaches 77.7–95.0%. This shows a clear gap from the baselines, which hold direction information only by chance or not at all in their structure (§6). Third, we can translate this gap into the language of the scenario structure that economic-value-based valuation (ICS / IFRS 17, and ESR in Japan) requires. The explicit output of the improvement rate $i^*(x, y)$, whose direction was tested in advance, can generate level, trend, and catastrophe shocks, sensitivity disclosures, and ORSA stress in an internally consistent way. It only needs setting changes that share the same fit (§7). The baseline methods that win on point forecasts cannot play this role because of their structure (§7.3). In short, the true value of Scale BB-D is not a replacement for point forecasts. It is a scenario generator.

The BEL sensitivity demo in §8 gave numbers for the practical meaning of this repositioning. The calculation followed the ESR implementation details (the risk-free interest rate curve and the Pillar 1 stress factors set by regulatory notice). Under the uniform level shock in the regulation (incidence rate +20%), the effect on the present value of benefits is almost constant across entry ages (+17.2 to 19.1%). In contrast, the trend shock (improvement rate set to zero) has a clear age gradient: +31 to 32% for entry age 30 and +10 to 11% for entry age 60. For young policies, it reaches about 1.7 times the regulatory level shock. The uncertainty of the improvement trend concentrates in young, long-term policies. The current practice of a static table with uniform factors cannot express this age structure of the risk in principle. The demo shows, in the vocabulary of the regulation, a direct link: whether an assumption base can handle trend scenarios decides the quality of risk understanding for a portfolio.

The contribution of this study has two layers. As novelty A, we extended a Scale BB type improvement scale to cause-specific mortality. As far as the authors know, this is the first system that separates and extracts improvement trends, period shocks, and cohort effects for each cause of death in one framework. As novelty B, we showed that this output becomes a direct assumption, without any proxy, for death benefits contingent on specific diseases — products where cause-specific death is the benefit trigger. So we set up two layers of application targets: medical insurance (proxy application to incidence rates) and death benefits contingent on specific diseases (direct application). For the former, the direction test in §6 works as a check of the methodology. For the latter, it works as a check of the assumption itself.

This year, ICS came into force and ESR started to apply in Japan. It is a turning point: assumption practice for disease risk must move from “a single point forecast” to “a consistent set of scenarios”. This study showed that the tool that meets this demand is not a new predictor that wins the accuracy race. The right tool is a framework that first fixes, as its base, a property we can test even with limited data — the direction of improvement rates. It then turns long-term assumptions, which tests cannot reach, into scenarios through explicit parameters that people can interpret. The extension of a “provisional scale”, which loses to simple methods in point forecasts, has a structural advantage in the era of economic-value-based valuation. This paradox is the new finding that this paper found beyond the negative conclusion of the backtest.